Chapter 7

BASIC OP-AMP CIRCUITS

CHAPTER OUTLINE

7-1 Comparators

7-2 Summing Amplifiers

7-3 Integrators and Differentiators

7-4 Active Diode Circuits

7-5 Troubleshooting

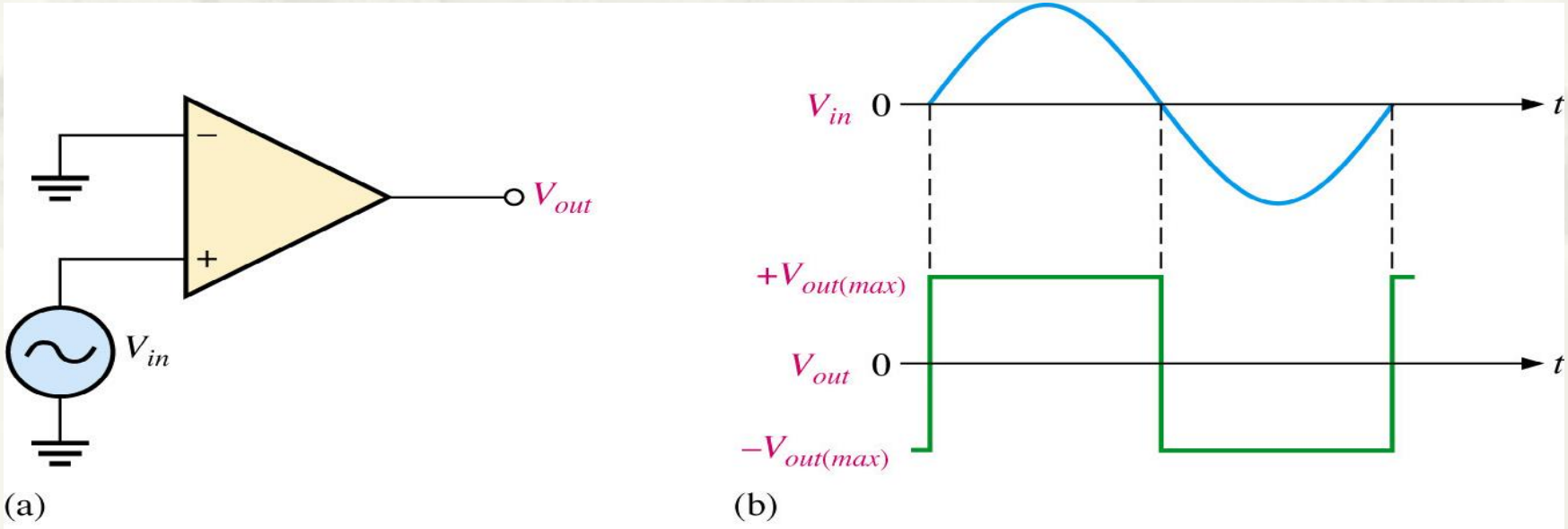
Fundamental applications of op-amps

7-1 Comparators

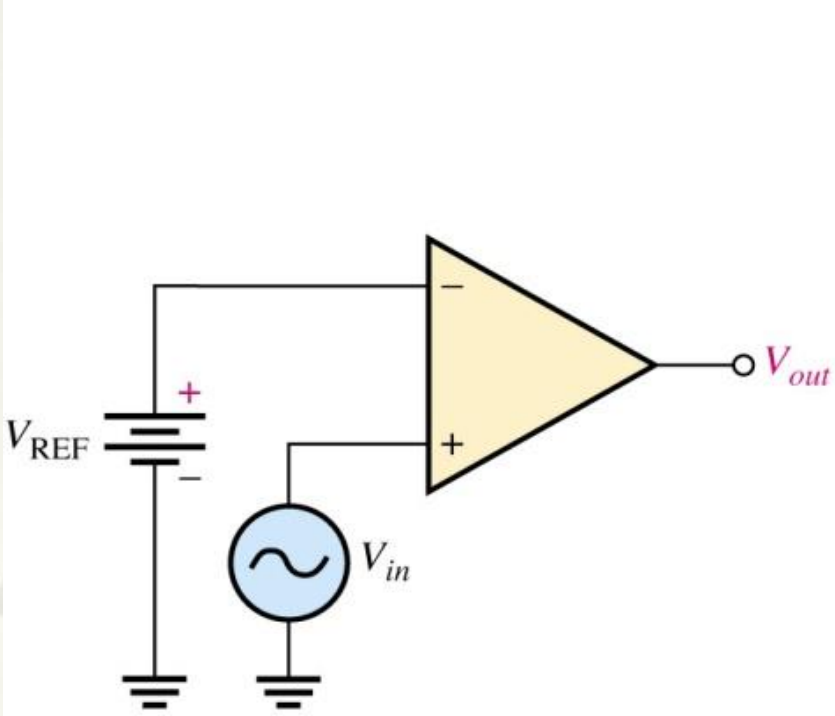
- Zero-Level Detector
- Nonzero-Level Detection
- Effects of Input Noise on Comparator Operation
- Reducing Noise Effects with Hysteresis
- A Comparator Application: Over-Temperature Sensing Circuit
- Op-Amp VS. Comparator

Zero-Level Detector

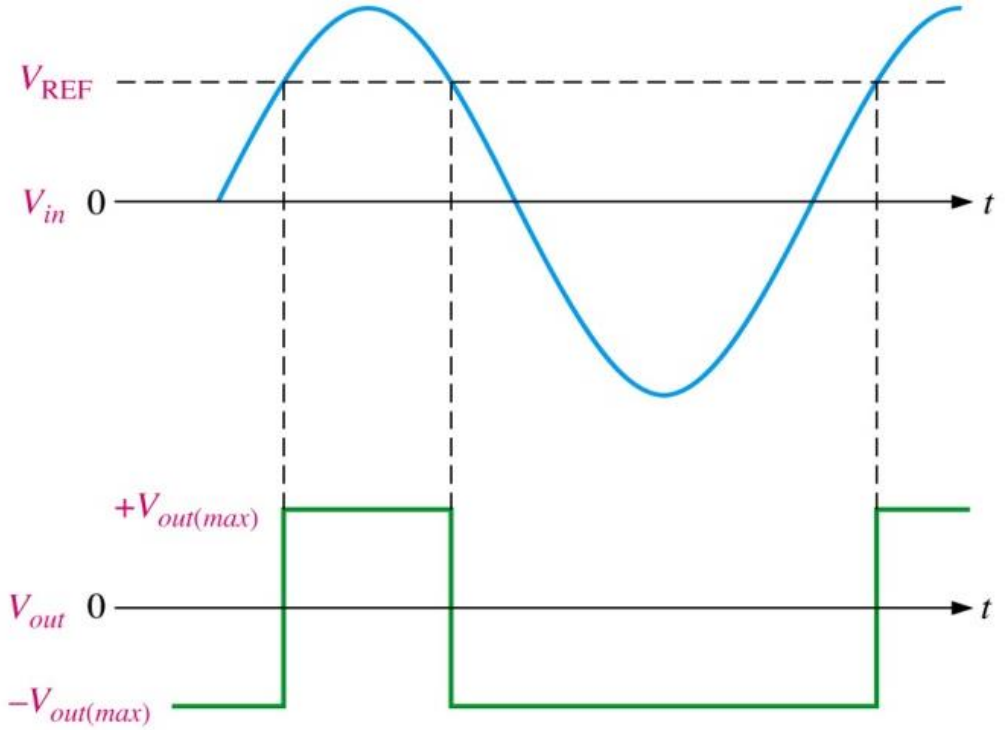
- **Comparator**
  - A comparator is a circuit that compares two input voltages and produces an output in either of two states indicating the greater than or less than relationship of the inputs.



Nonzero-Level Detection

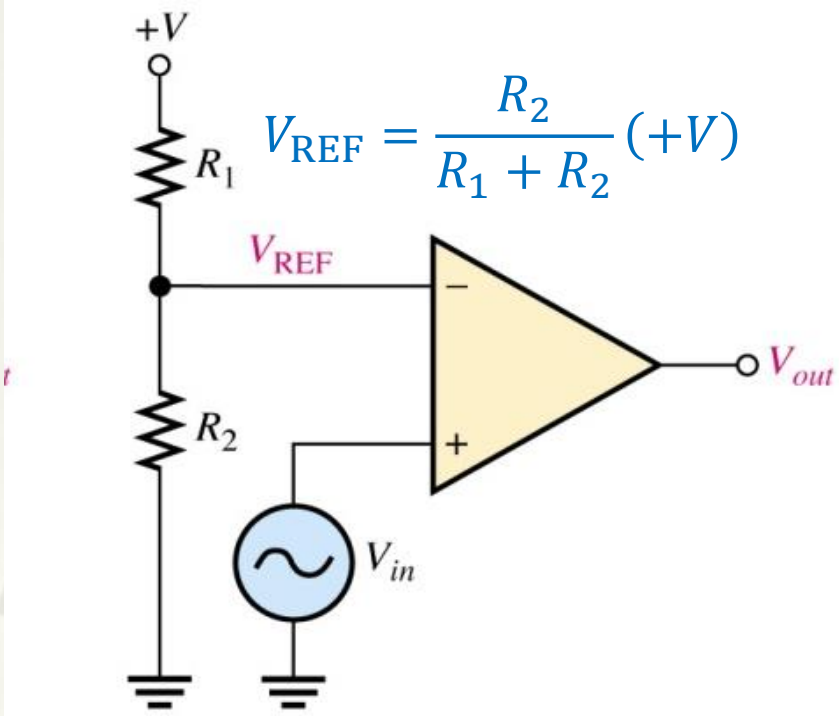


(a) Battery reference



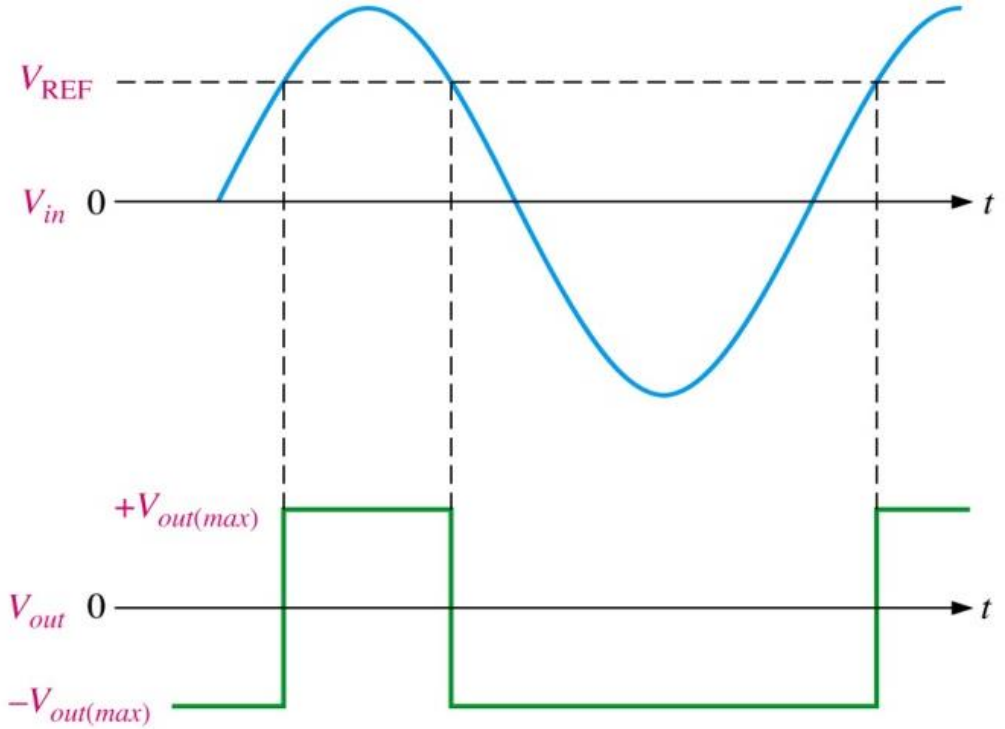
(d) Waveforms

Nonzero-Level Detection



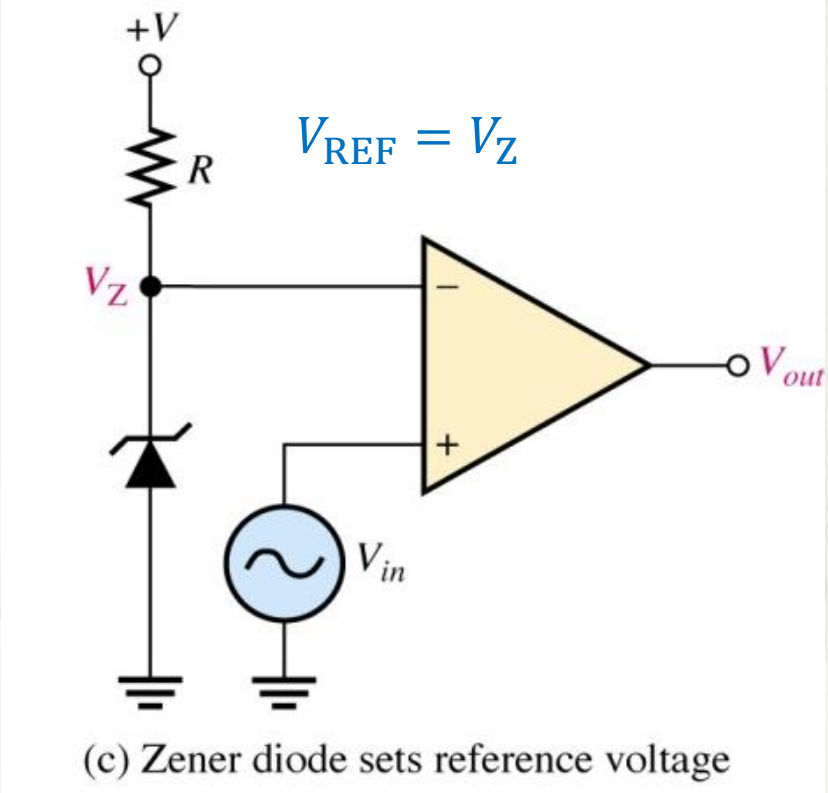
$$V_{\text{REF}} = \frac{R_2}{R_1 + R_2}(+V)$$

(b) Voltage-divider reference

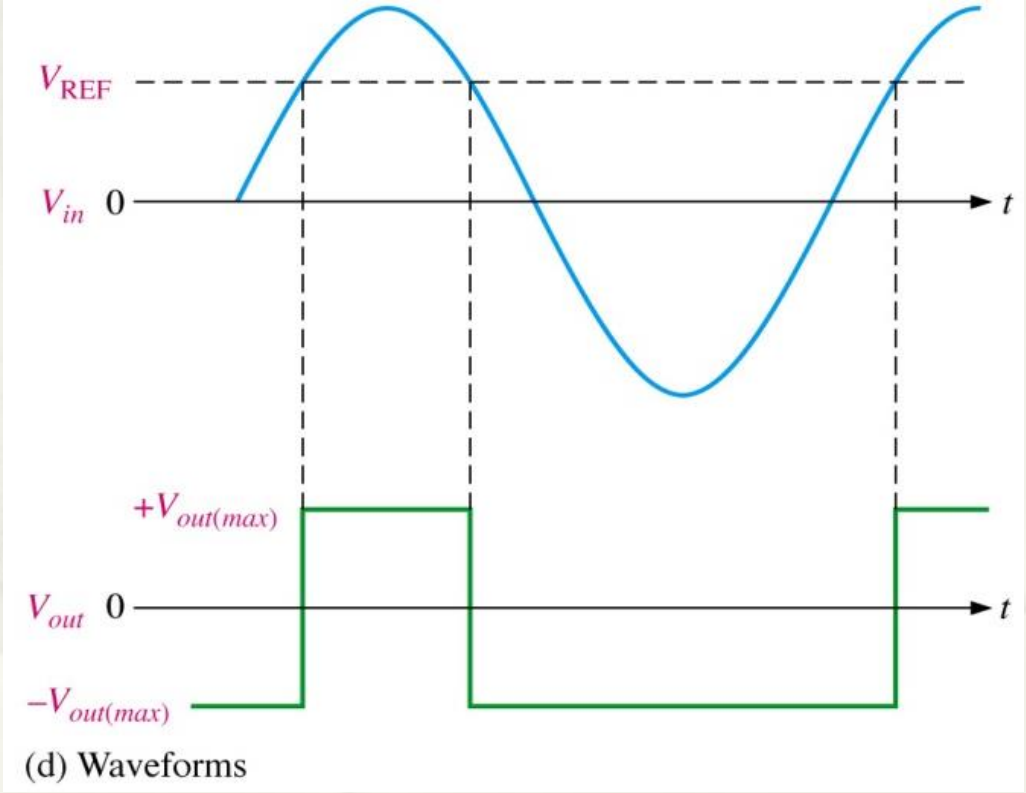


(d) Waveforms

Nonzero-Level Detection



(c) Zener diode sets reference voltage

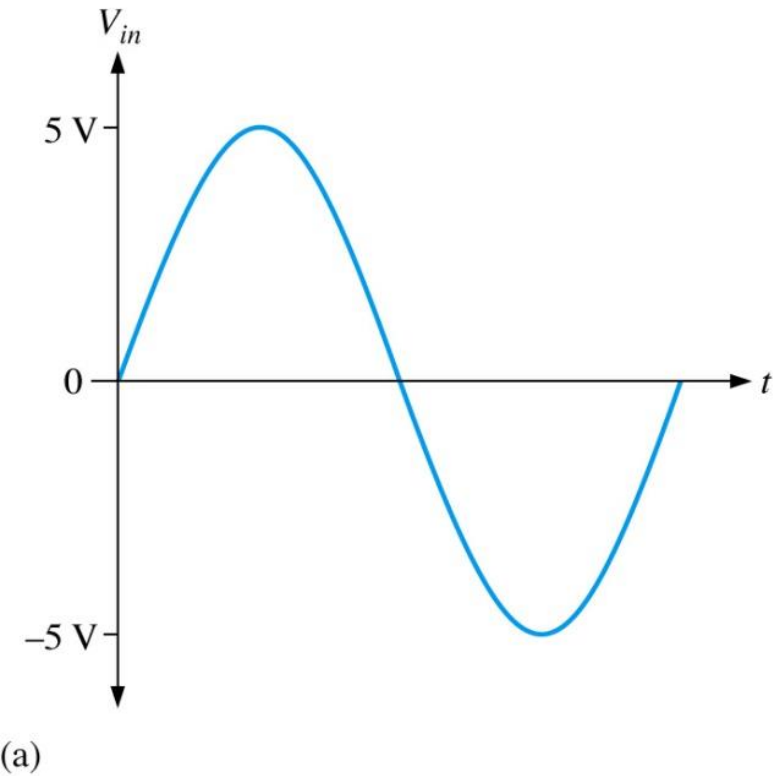


(d) Waveforms

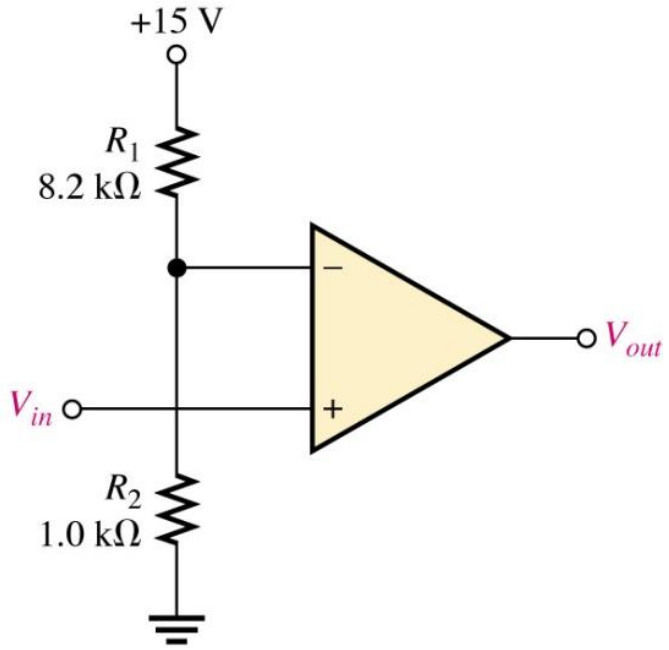
Example 7-1

Problem

The input signal in Figure 7–3(a) is applied to the comparator circuit in Figure 7–3(b). Make a sketch of the output showing its proper relationship to the input signal. Assume the maximum output levels of the op-amp are ±12 V.



(a)



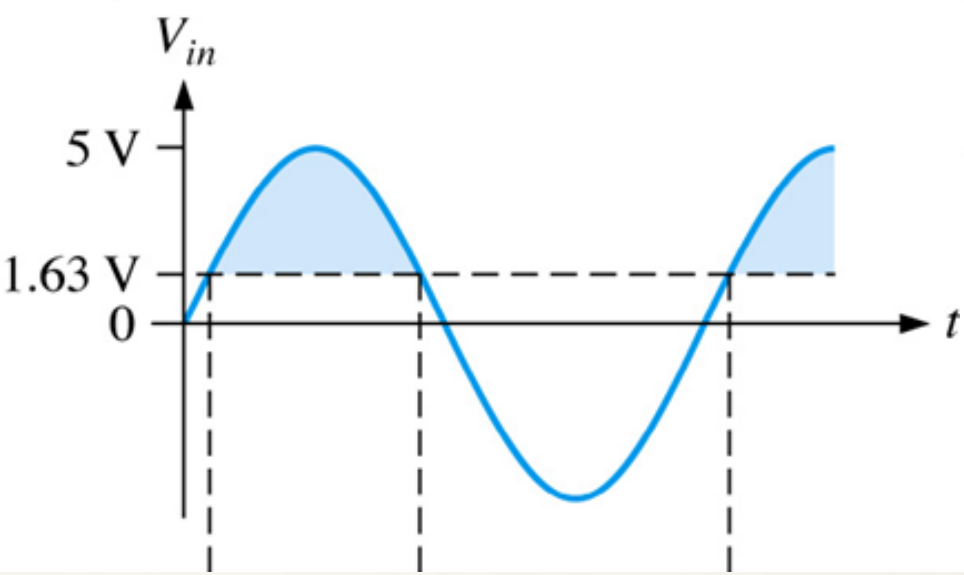
(b)

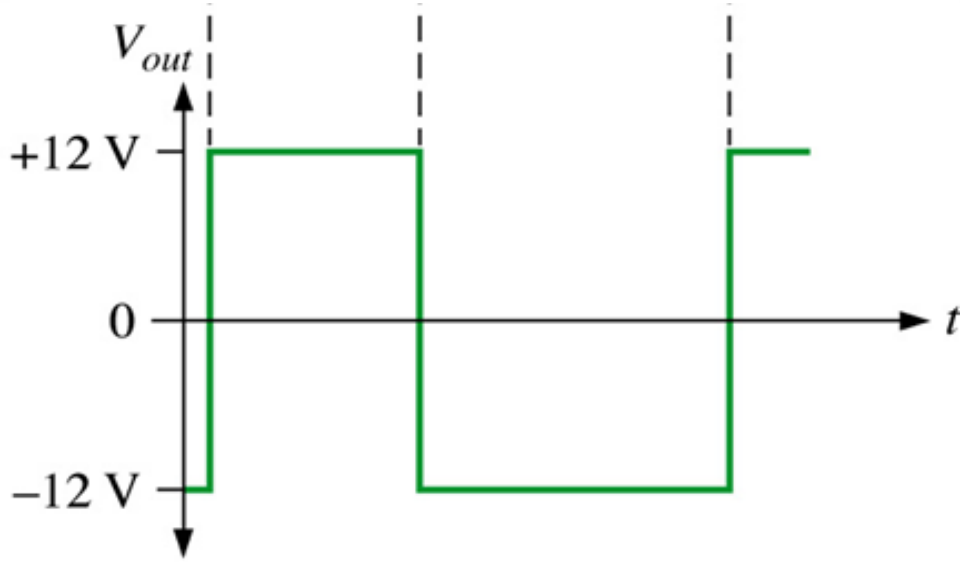
Example 7-1

Solution

The reference voltage is set by R_1 and R_2 as follows:

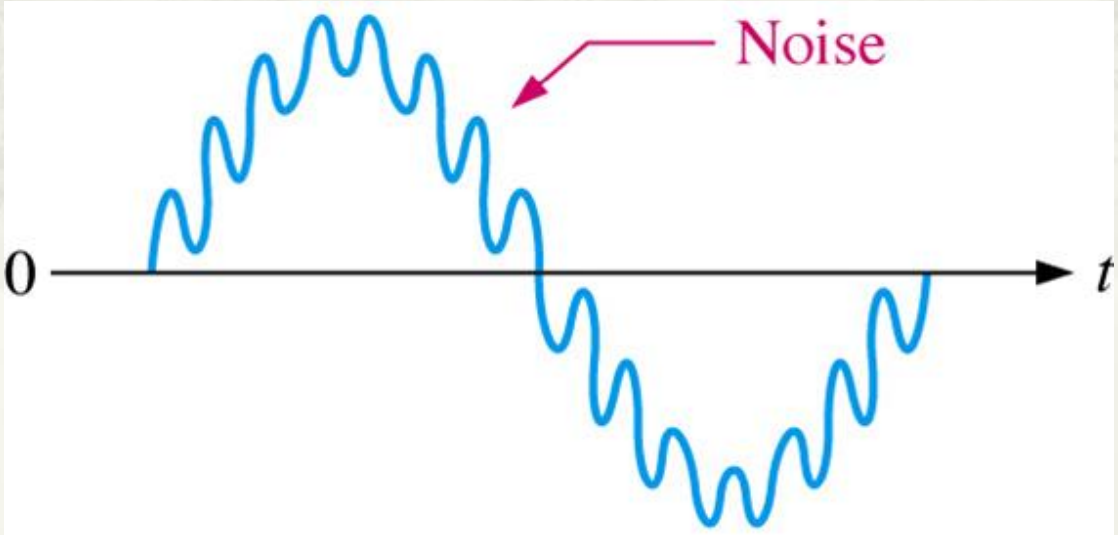
$$V_{REF} = \frac{R_2}{R_1 + R_2}(+V) = \frac{1.0\,\text{k}\Omega}{8.2\,\text{k}\Omega + 1.0\,\text{k}\Omega}(+15\text{ V}) = 1.63\text{ V}$$



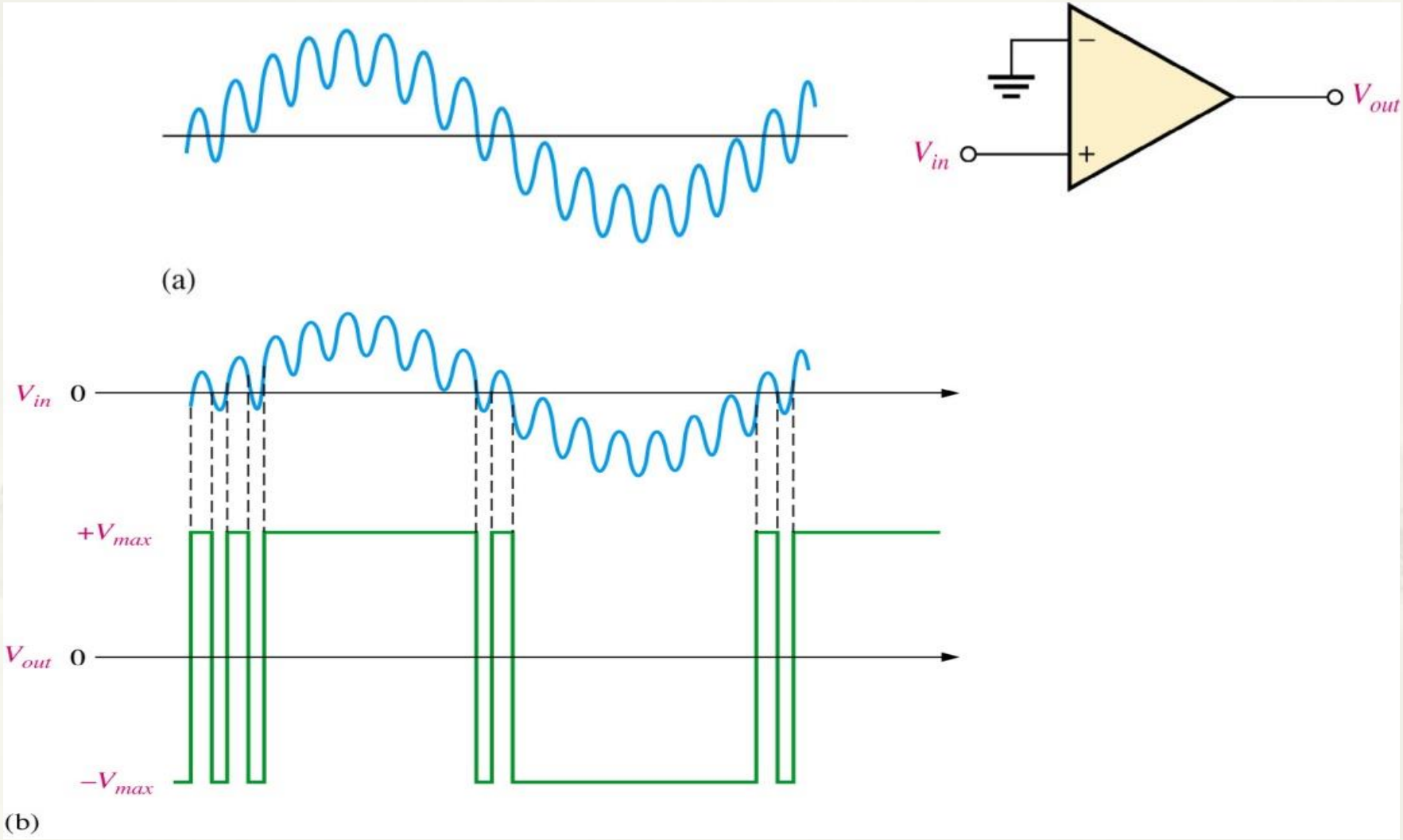


Effects of Input Noise on Comparator Operation

- In many practical situations, noise (unwanted voltage or current fluctuations) may appear on the input line.
- The noise voltage becomes superimposed on the input voltage, and can cause a comparator to erratically switch output states.



Effects of Input Noise on Comparator Operation



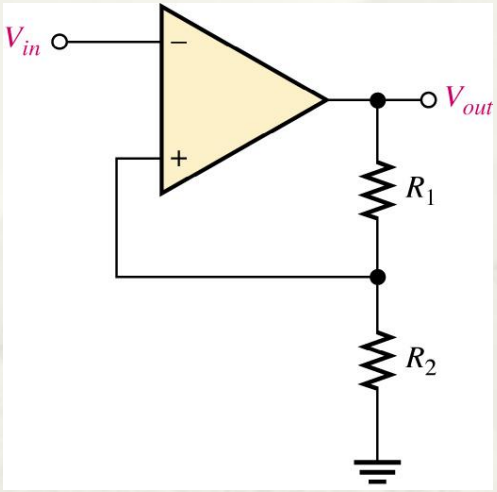
Reducing Noise Effects with Hysteresis

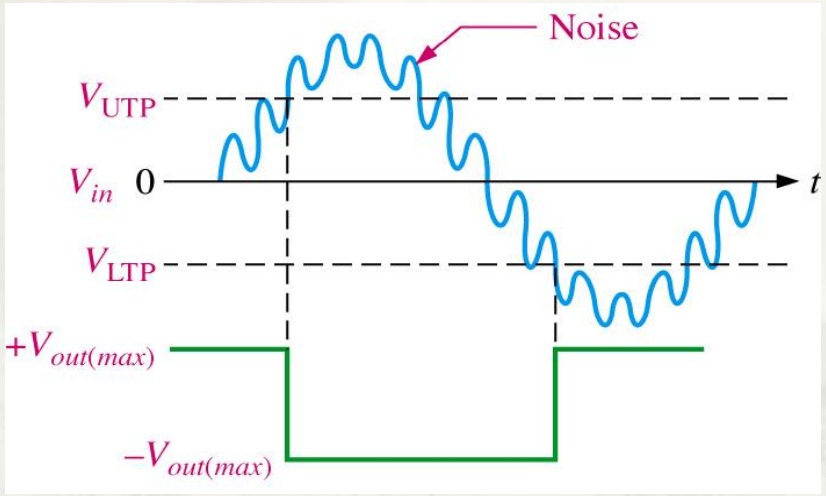
- **Comparator with positive feedback for hysteresis**
  - In order to make the comparator less sensitive to noise, a technique incorporating positive feedback, called hysteresis, can be used.



Reducing Noise Effects with Hysteresis

- **Comparator with positive feedback for hysteresis**





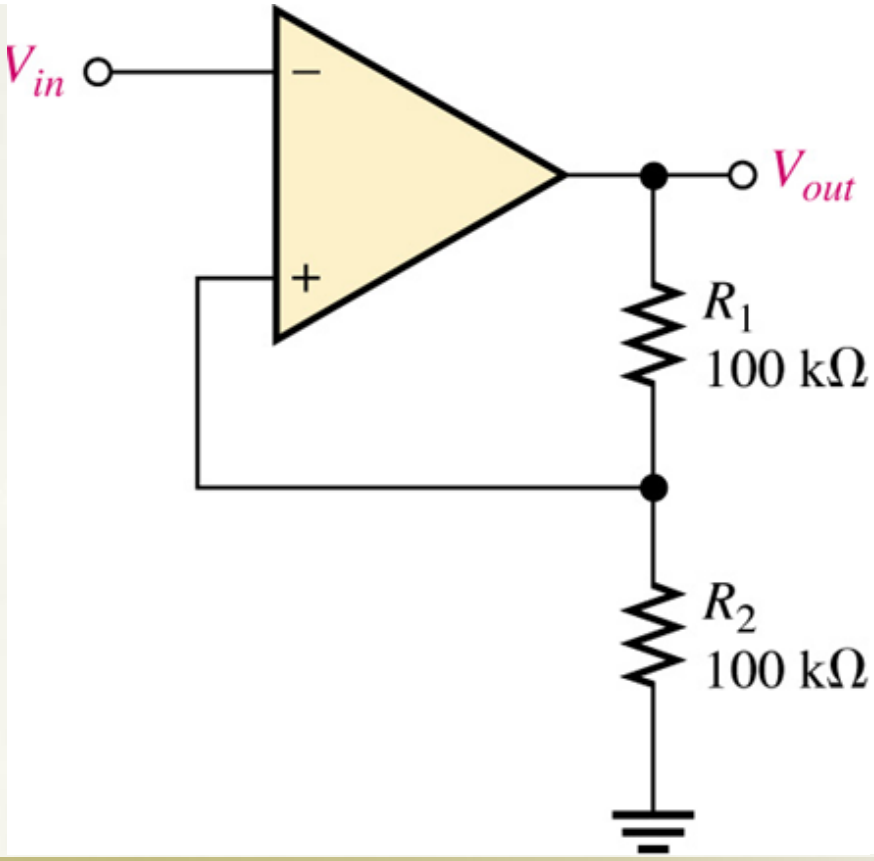
- The upper trigger point (UTP)
  - ✓ $V_{\text{UTP}} = \left(\frac{R_2}{R_1 + R_2}\right)\left(+V_{out(max)}\right)$
- The lower trigger point (LTP)
  - ✓ $V_{\text{LTP}} = \left(\frac{R_2}{R_1 + R_2}\right)\left(-V_{out(max)}\right)$

A comparator with hysteresis is sometimes known as a **Schmitt trigger**.

Example 7-2

Problem

Determine the upper and lower trigger points and the hysteresis for the comparator circuit in Figure 7–9. Assume that $+V_{out(max)} = +5$ V and $-V_{out(max)} = -5$ V.



Example 7-2

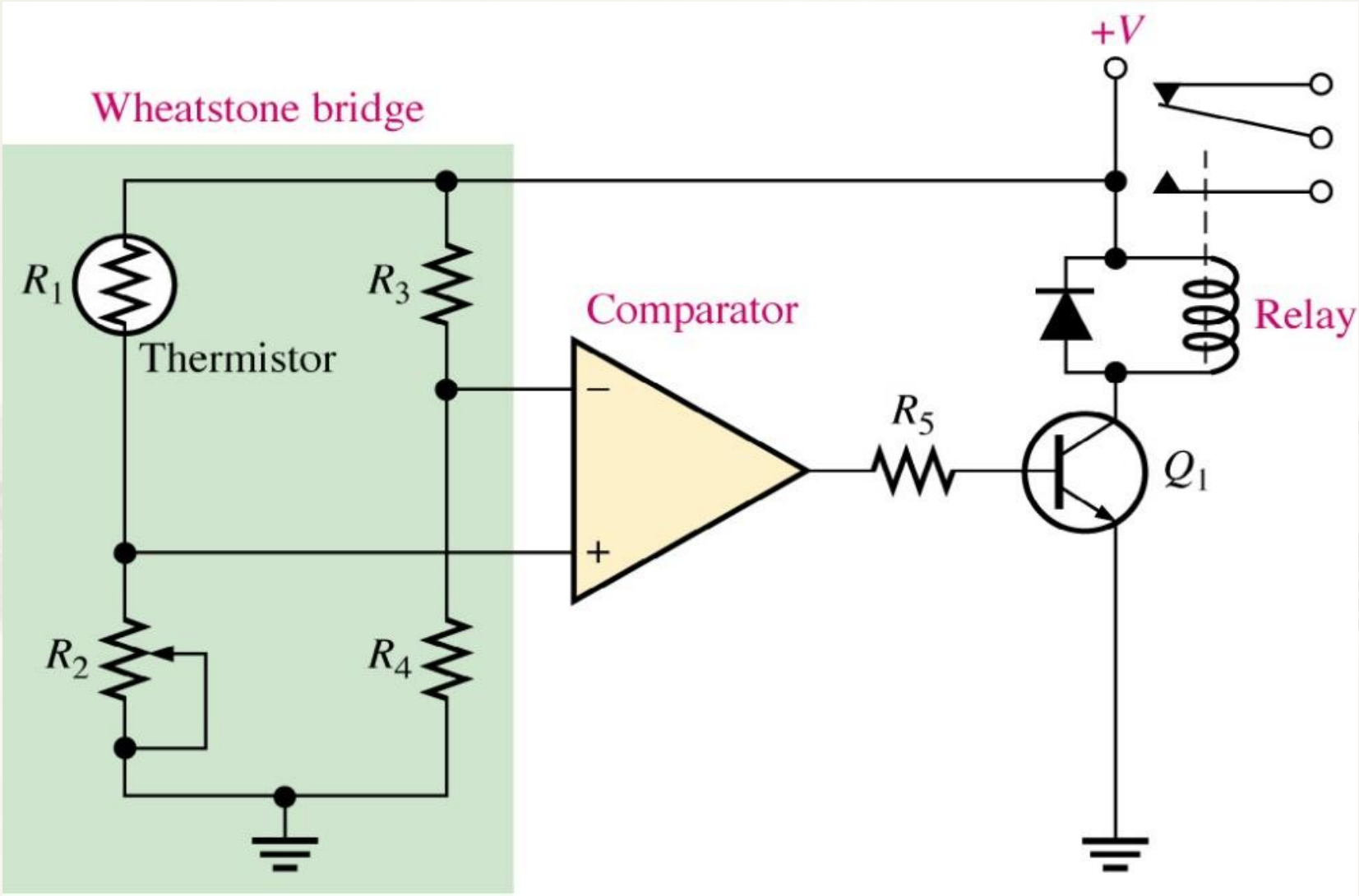
Solution

$$V_{UTP} = \frac{R_2}{R_1 + R_2}(+V_{out(max)}) = 0.5(5\text{ V}) = +\mathbf{2.5\text{ V}}$$

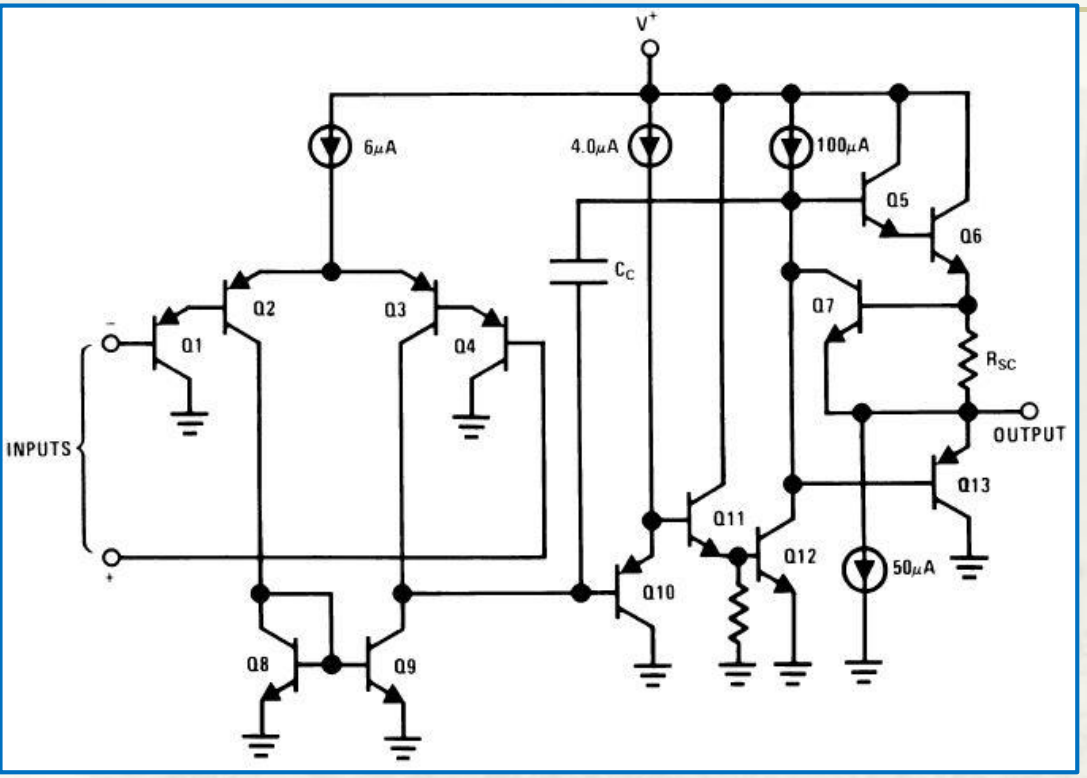
$$V_{LTP} = \frac{R_2}{R_1 + R_2}(-V_{out(max)}) = 0.5(-5\text{ V}) = -\mathbf{2.5\text{ V}}$$

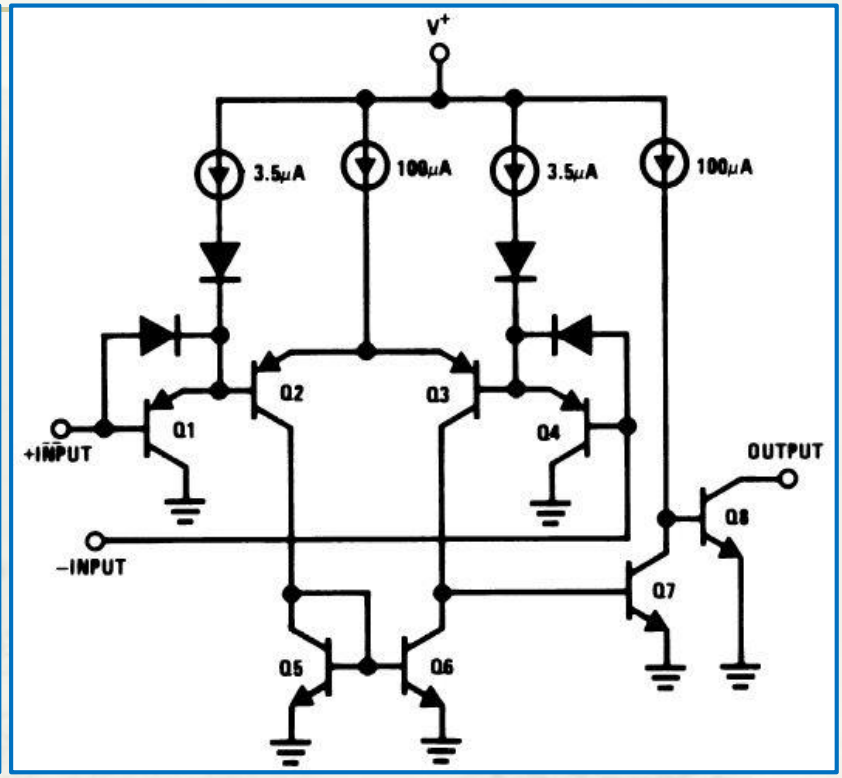
$$V_{HYS} = V_{UTP} - V_{LTP} = 2.5\text{ V} - (-2.5\text{ V}) = \mathbf{5\text{ V}}$$

A Comparator Application: Over-Temperature Sensing Circuit



Op-Amp VS. Comparator





Op-amp LM358

- Speed: relatively low
- Output stage: push-pull
- Linearity: good

Comparator LM393

- Speed: high
- Output stage: OC
- Linearity: bad

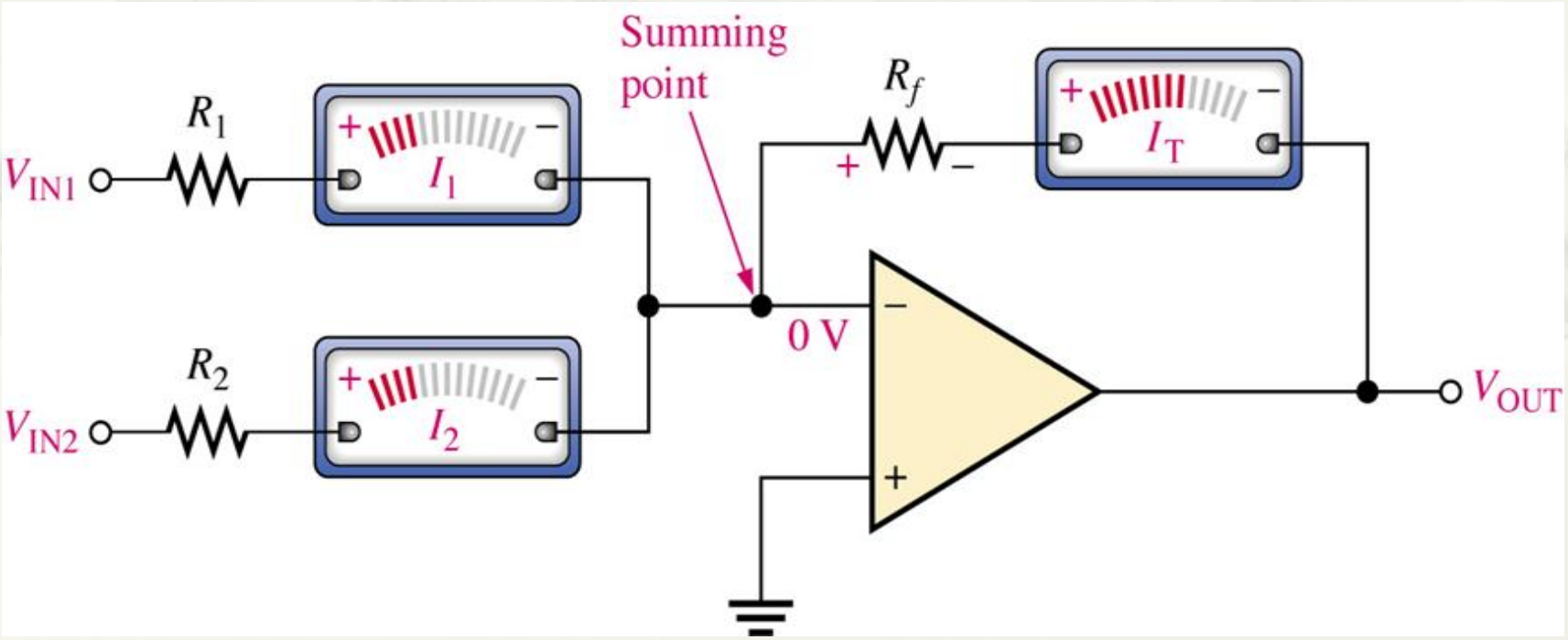
7-2 Summing Amplifiers

- Averaging Amplifier
- Scaling Adder
- A Scaling Adder Application:

  Digital-to-Analog (D/A) Conversion

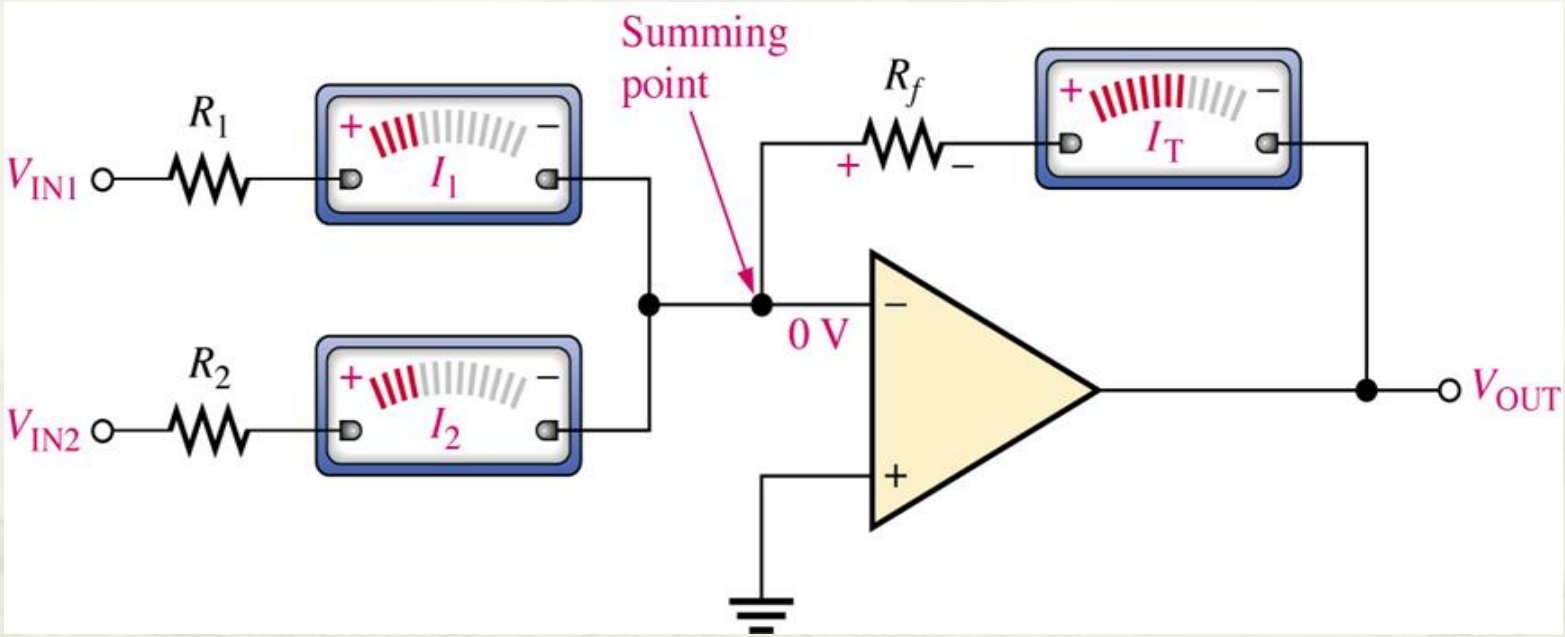
Summing Amplifiers

- **Summing amplifier**
  - The summing amplifier has two or more inputs, and its output voltage is proportional to the negative of the algebraic sum of its input voltages.



Summing Amplifiers

- **Two-input summing amplifier**



$$I_T = I_1 + I_2$$

$$V_{OUT} = -I_T R_f = -(I_1 + I_2)R_f$$

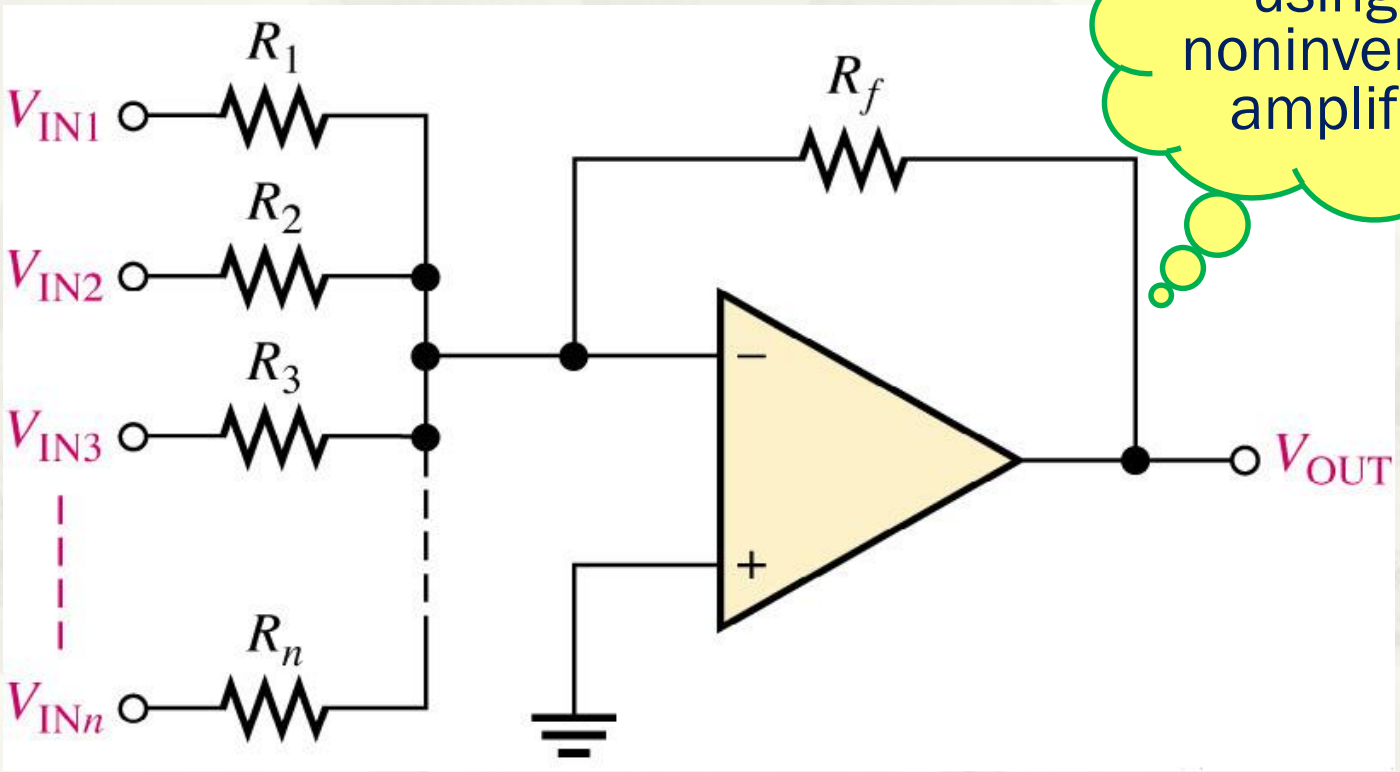
$$\boldsymbol{V_{OUT} = -\left(\frac{V_{IN1}}{R_1} + \frac{V_{IN1}}{R_2}\right)R_f}$$

If $R_1 = R_2 = R_f = R$

$$\boldsymbol{V_{OUT} = -(V_{IN1} + V_{IN2})}$$

Summing Amplifiers

- **Summing amplifier with *n* inputs**

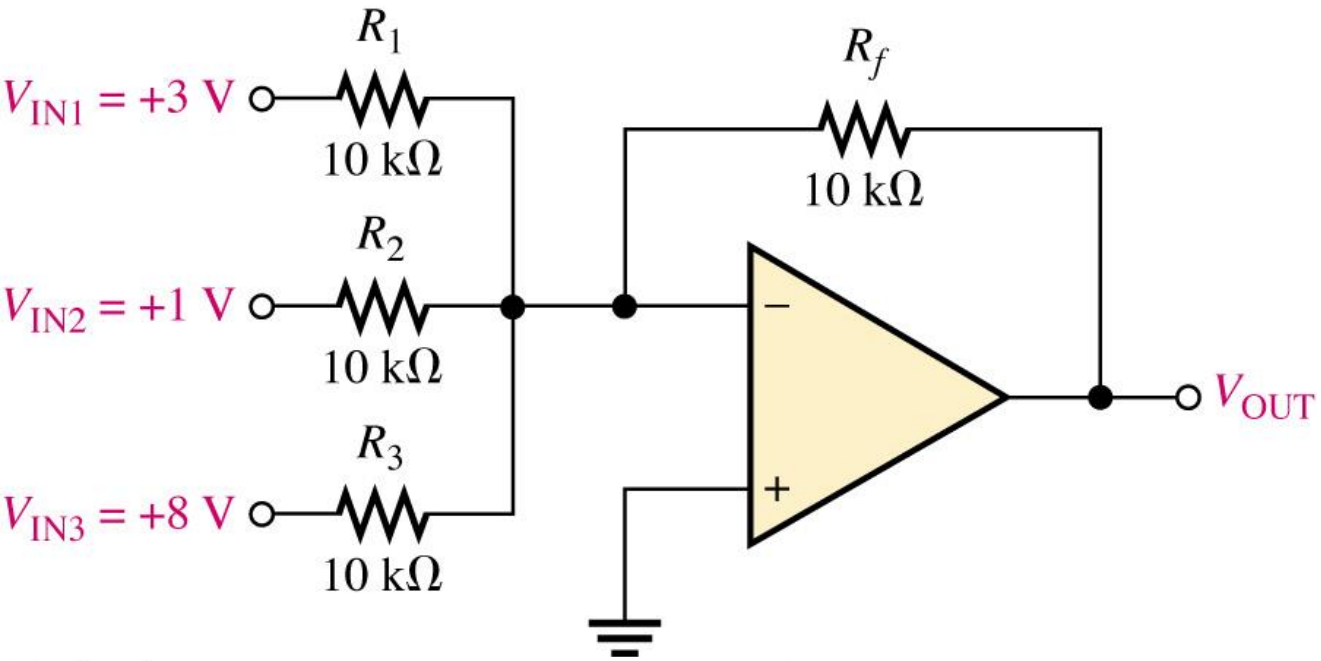


$$V_{OUT} = -(V_{IN1} + V_{IN2} + \cdots + V_{INn})$$

Example 7-3

Problem

Determine the output voltage in Figure 7–14.



Solution

$$V_{OUT} = -(V_{IN1} + V_{IN2} + V_{IN3}) = -(3\text{ V} + 1\text{ V} + 8\text{ V}) = \mathbf{-12\text{ V}}$$

Averaging Amplifier

- A summing amplifier can be made to produce the mathematical average of the input voltages.

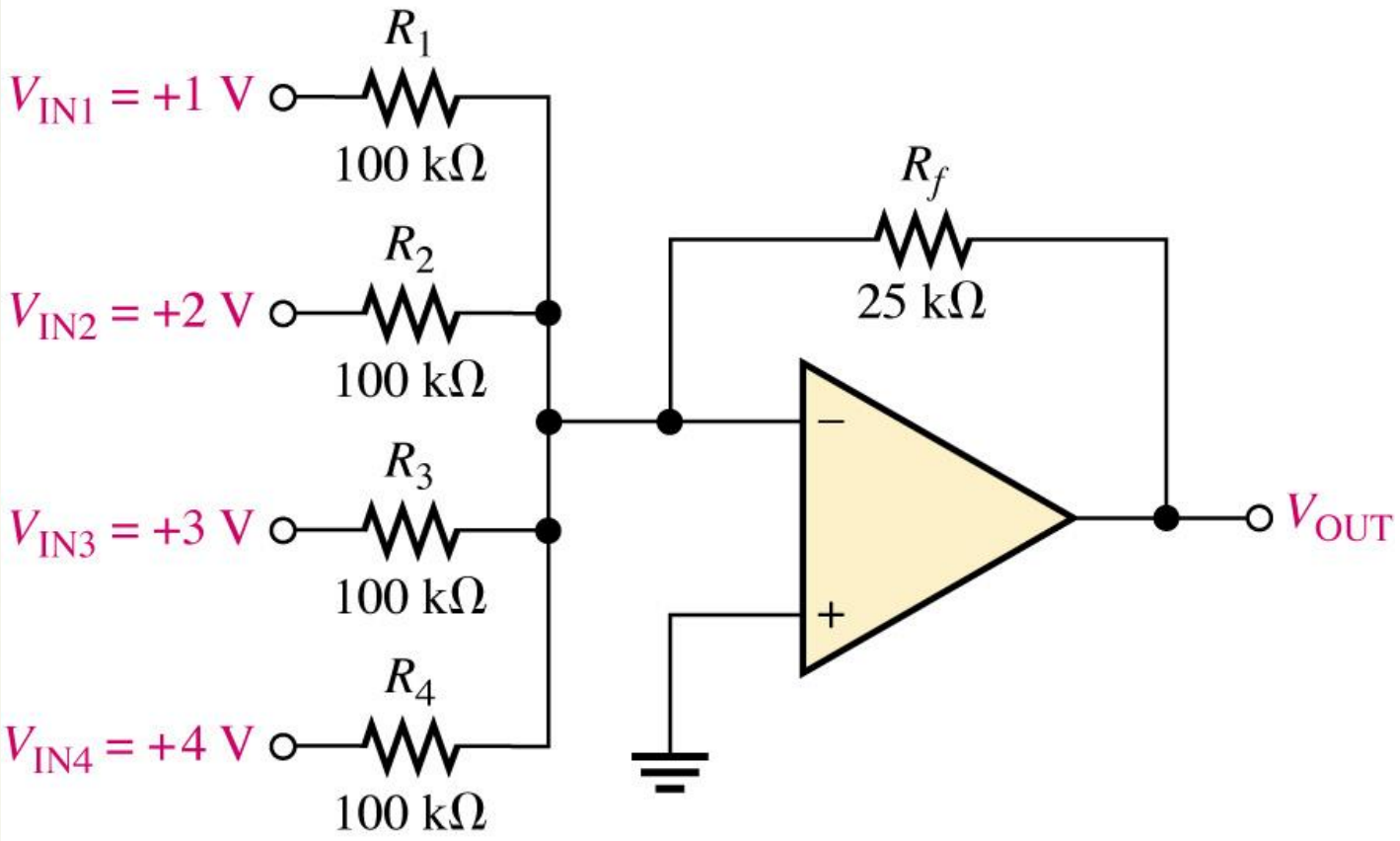
$$V_{\text{OUT}} = -\frac{R_f}{R}(V_{\text{IN1}} + V_{\text{IN2}} + \cdots + V_{\text{IN}n})$$

Averaging is done by setting $R_f / R = 1/n$, where n is the number of inputs.

Example 7-4

Problem

Show that the amplifier in Figure 7–15 produces an output whose magnitude is the mathematical average of the input voltages.



Example 7-4

Solution

Since the input resistors are equal, $R = 100\ \text{k}\Omega$. The output voltage is

$$V_{OUT} = -\frac{R_f}{R}(V_{IN1} + V_{IN2} + V_{IN3} + V_{IN4})$$

$$= -\frac{25\ \text{k}\Omega}{100\ \text{k}\Omega}(1\ \text{V} + 2\ \text{V} + 3\ \text{V} + 4\ \text{V}) = -\frac{1}{4}(10\ \text{V}) = \mathbf{-2.5\ V}$$

A simple calculation shows that the average of the input values is the same magnitude as V_{OUT} but of opposite sign.

$$V_{IN(avg)} = \frac{1\ \text{V} + 2\ \text{V} + 3\ \text{V} + 4\ \text{V}}{4} = \frac{10\ \text{V}}{4} = 2.5\ \text{V}$$

Scaling Adder

- A **different weight** can be assigned to each input of a summing amplifier by simply adjusting the values of the input resistors.

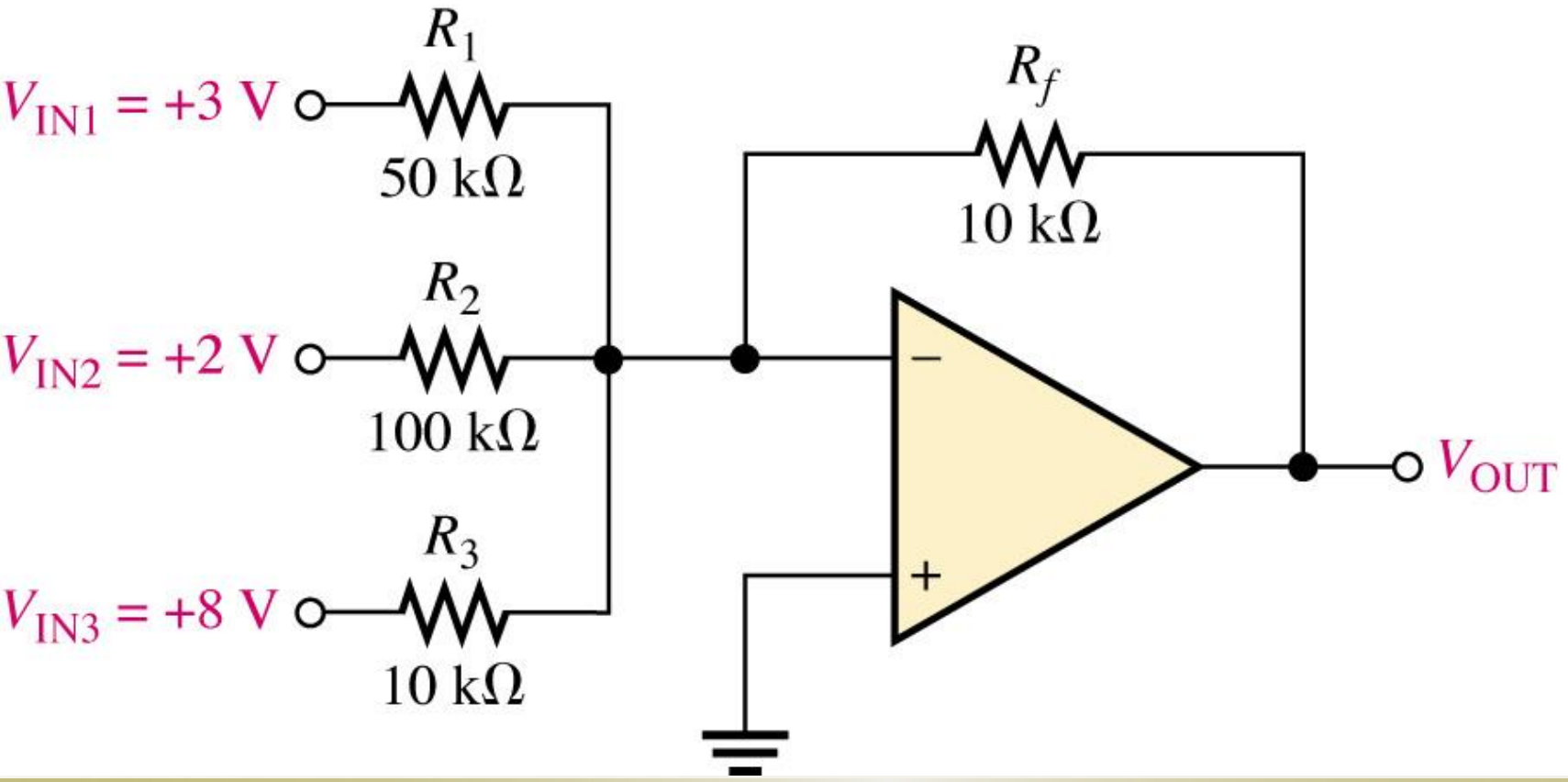
$$V_{\text{OUT}} = -\left(\frac{R_f}{R_1}V_{\text{IN1}} + \frac{R_f}{R_2}V_{\text{IN2}} + \cdots + \frac{R_f}{R_n}V_{\text{INn}}\right)$$

The weight of a particular input is set by the ratio of R_f to the resistance for that input.

Example 7-5

Problem

Determine the weight of each input voltage for the scaling adder in Figure 7–16 and find the output voltage.



Example 7-5

Solution

Weight of input 1: $\frac{R_f}{R_1} = \frac{10\,\text{k}\Omega}{50\,\text{k}\Omega} = \mathbf{0.2}$

Weight of input 2: $\frac{R_f}{R_2} = \frac{10\,\text{k}\Omega}{100\,\text{k}\Omega} = \mathbf{0.1}$

Weight of input 3: $\frac{R_f}{R_3} = \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega} = \mathbf{1}$

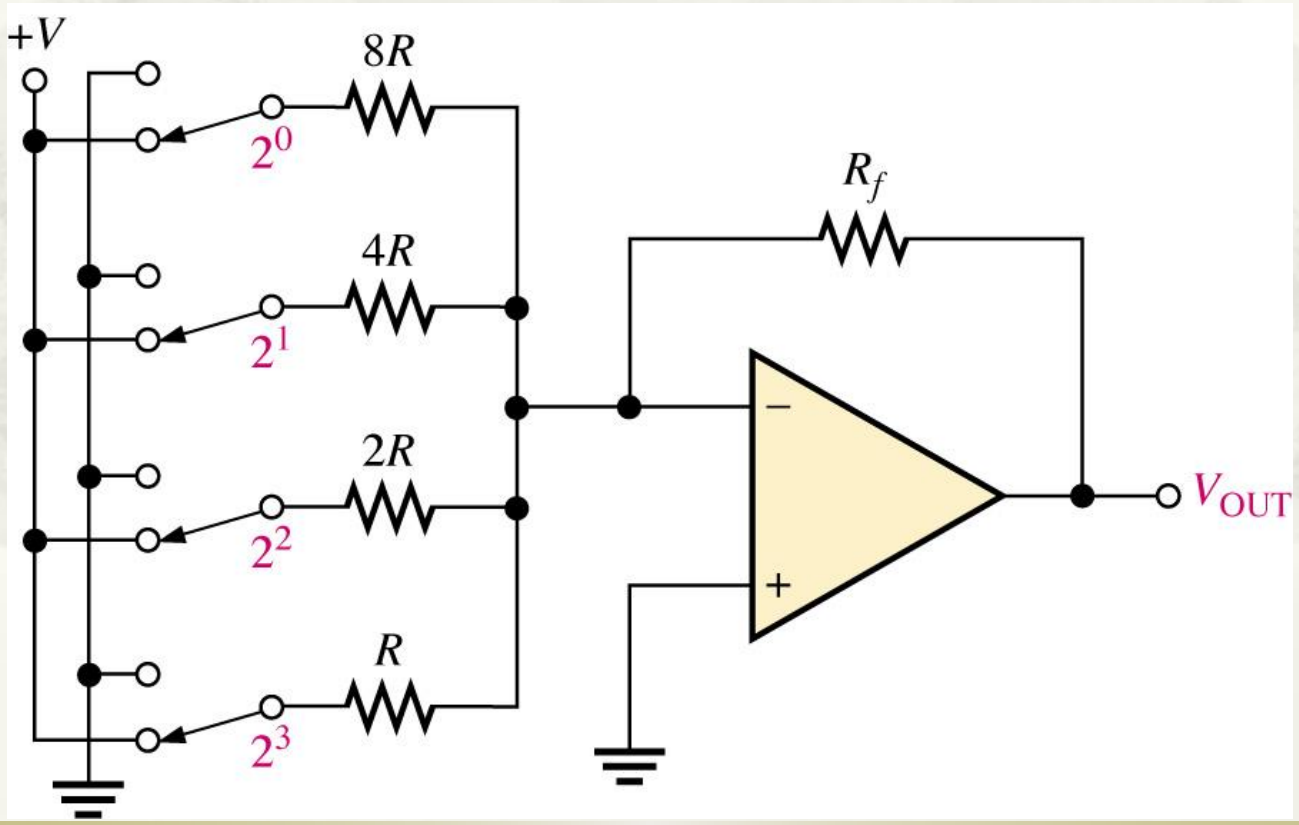
The output voltage is

$$V_{\text{OUT}} = -\left(\frac{R_f}{R_1}V_{\text{IN1}} + \frac{R_f}{R_2}V_{\text{IN2}} + \frac{R_f}{R_3}V_{\text{IN3}}\right)$$

$$= -[0.2(3\text{ V}) + 0.1(2\text{ V}) + 1(8\text{ V})] = -(0.6\text{ V} + 0.2\text{ V} + 8\text{ V}) = \mathbf{-8.8\text{ V}}$$

A Scaling Adder Application: Digital-to-Analog (D/A) Conversion

- **D/A conversion**
  - D/A conversion is an important interface process for converting digital signals to analog (linear) signals.

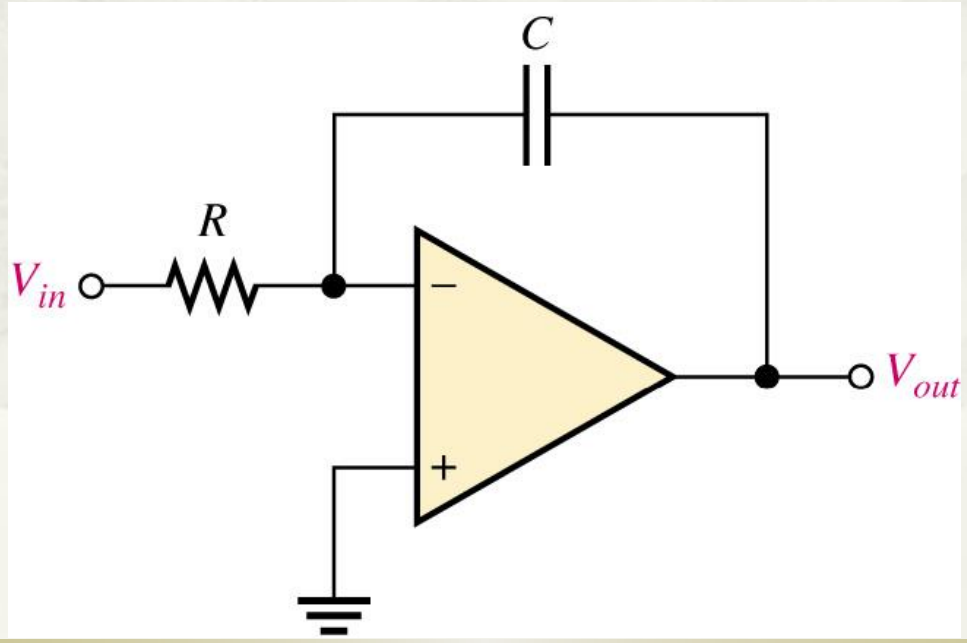


7-3 Integrators and Differentiators

- The Op-Amp Integrator
- The Op-Amp Differentiator

The Op-Amp Integrator

- **Integrator**
  - An integrator is a circuit that produces an inverted output that approximates the area under the curve of the input function.
- **Ideal integrator**



The Op-Amp Integrator

What about in a simple *RC* circuit?

How a Capacitor Charges

$$Q = I_C t$$
$$Q = CV_C$$

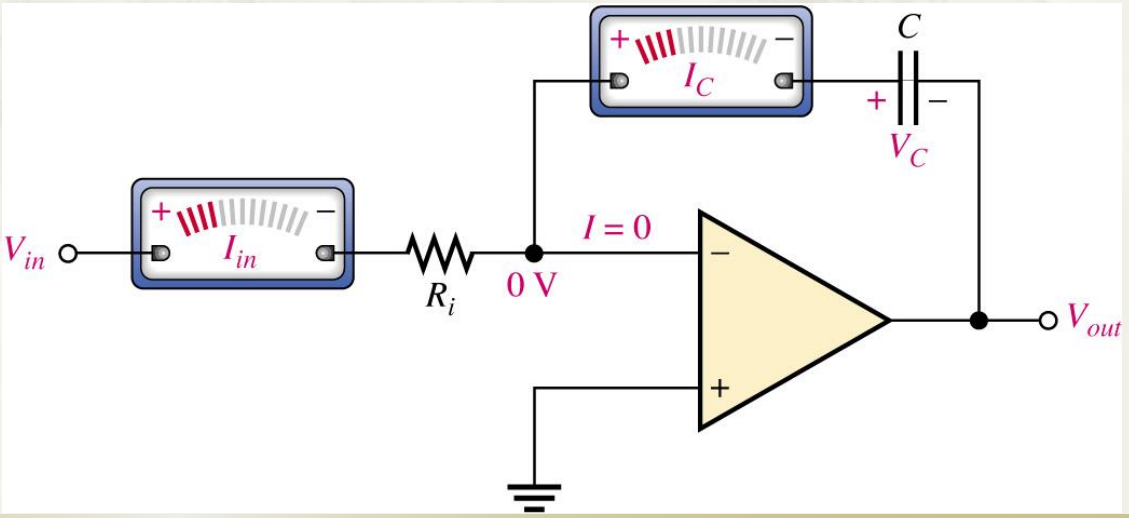
$$V_C = \left(\frac{I_C}{C}\right) t$$
$$I_{in} = \frac{V_{in}}{R_i}$$
$$I_C = I_{in}$$

For ac value:

$$V_C = \frac{V_{in}}{R_i C} \cdot t$$

For instantaneous value:

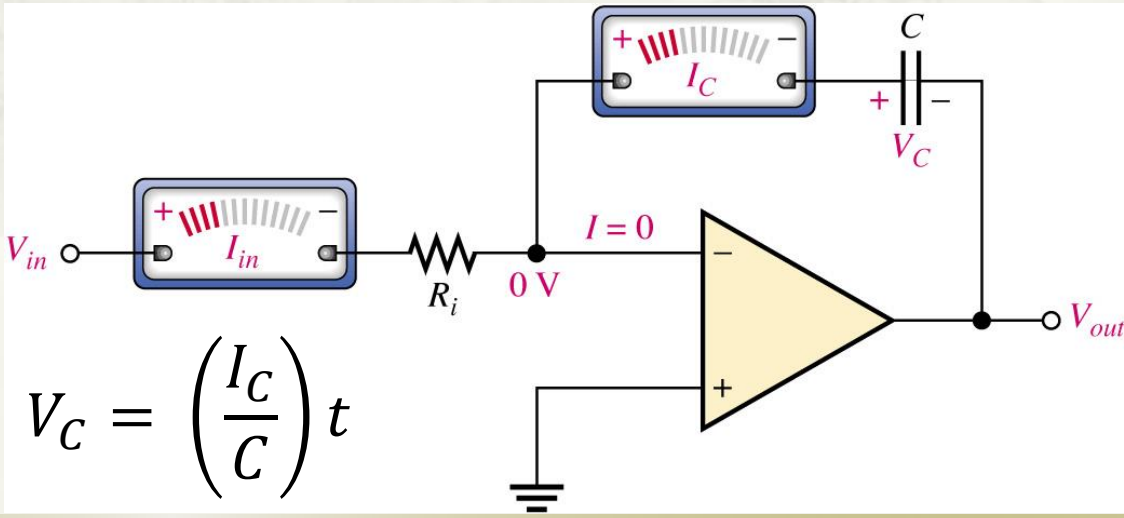
$$v_C = \frac{1}{R_i C} \int v_{in} dt$$



The Op-Amp Integrator

- **In a simple *RC* circuit**
  - The capacitor voltage is not linear but is exponential.
  - The charging current continuously decrease as the capacitor charges and causes the rate of charge of the voltage to continuously decrease.
- **The advantage of op-amp integrator**

The capacitor's charging current is made constant, thus producing a straight-line (linear) voltage rather than an exponential voltage.



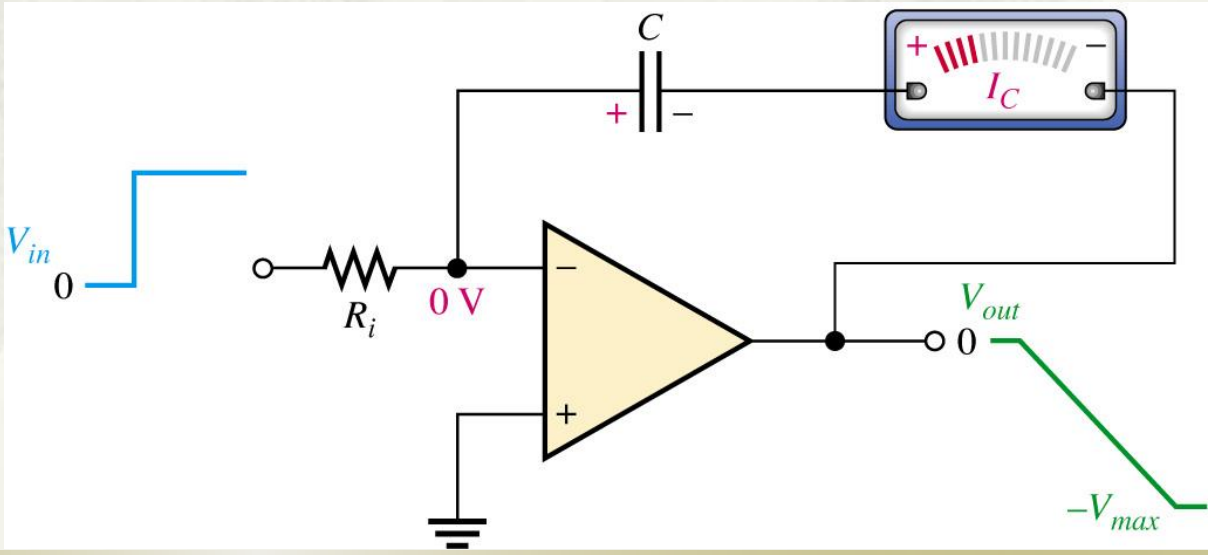
$$V_C = \left(\frac{I_C}{C}\right) t$$

The Op-Amp Integrator

- **The Capacitor Voltage**
  - Since V_{in} is constant, the voltage on the negative side of the capacitor decreases linearly from zero as the capacitor charges. This voltage is called a negative ramp.
- **The Output Voltage**
  - V_{out} is the same as the voltage on the negative side of the capacitor.

$$V_{out} = -\frac{V_{in}}{R_i C} \cdot t$$

$$v_{out} = -\frac{1}{R_i C}\int v_{in} dt$$

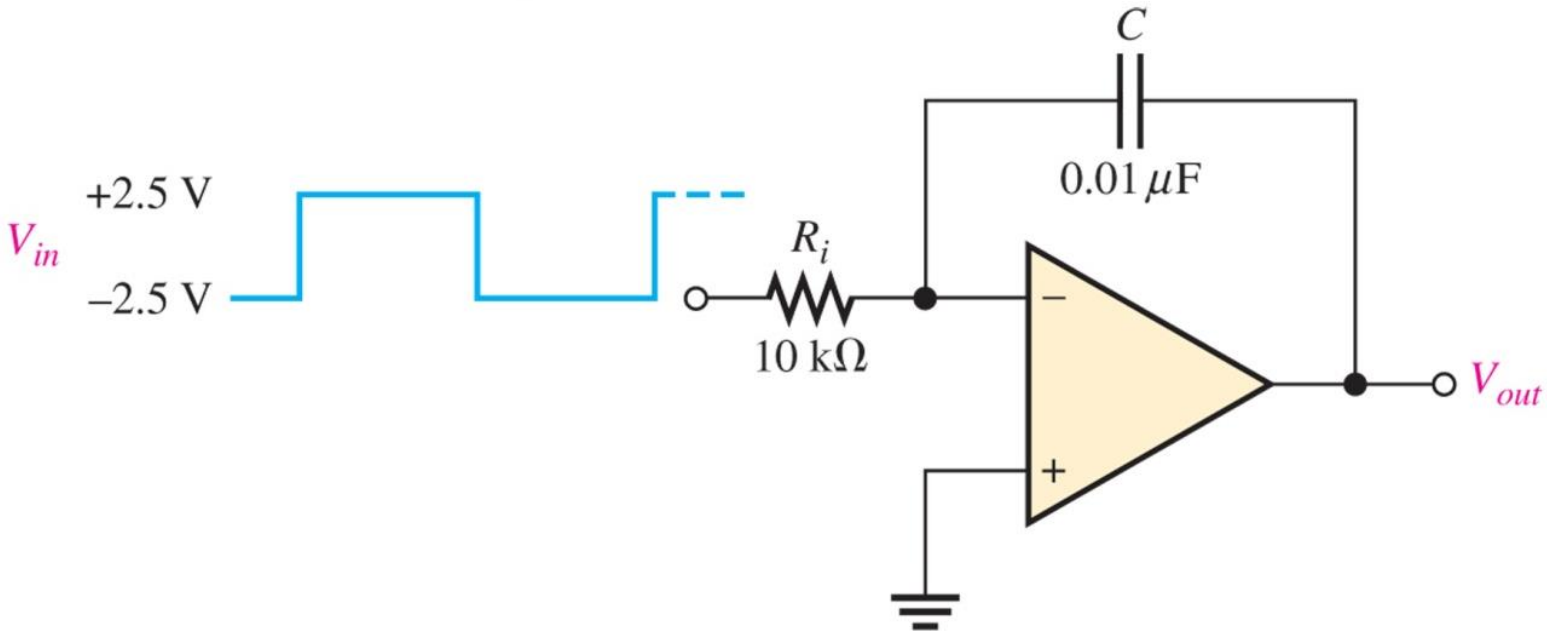


Example 7-6

Problem

(a) Determine the rate of change of the output voltage in response to the input square wave, as shown for the ideal integrator in Figure 7–22(a). The output voltage is initially zero. The pulse width is 100 μs.

(b) Describe the output and draw the waveform.



(a)

Example 7-6

Solution

(a) During the time that the input is positive, the capacitor is charging such that the rate of change of the output voltage is negative.

$$\text{Output rate of change} = -\frac{V_{in}}{R_i C} = -\frac{5\text{ V}}{(10\text{ k}\Omega)(0.01\ \mu\text{F})} = -50\text{ kV/s}$$

$$= \mathbf{-50\ mV/\mu s}$$

During the time that the input is negative, the capacitor is discharging; so the rate of change of the output voltage is the same as during charging but it is positive.

$$\text{Output rate of change} = \mathbf{+50\ mV/\mu s}$$

Example 7-6

Solution

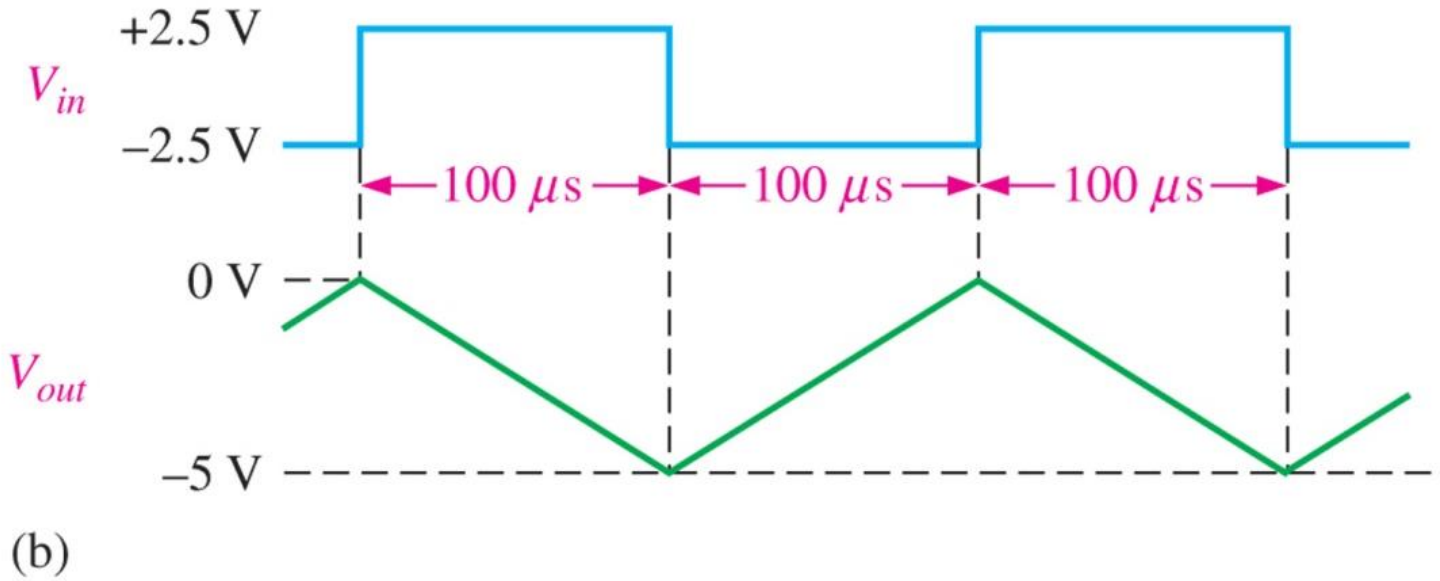
(b) When the input is at +2.5 V, the output is a negative-going ramp. When the input is at −2.5 V, the output is a positive-going ramp.

$$\text{Change in output voltage} = (50\ \text{mV}/\mu\text{s})(100\ \mu\text{s}) = 5\ \text{V}$$

During the time the input is at +2.5 V, the output will go from 0 V to − 5 V. During the time the input is at −2.5 V, the output will go from −5 V to 0 V. Therefore, the output is a triangular wave with peaks at 0 V and −5 V, as shown in Figure 7–22(b).

Example 7-6

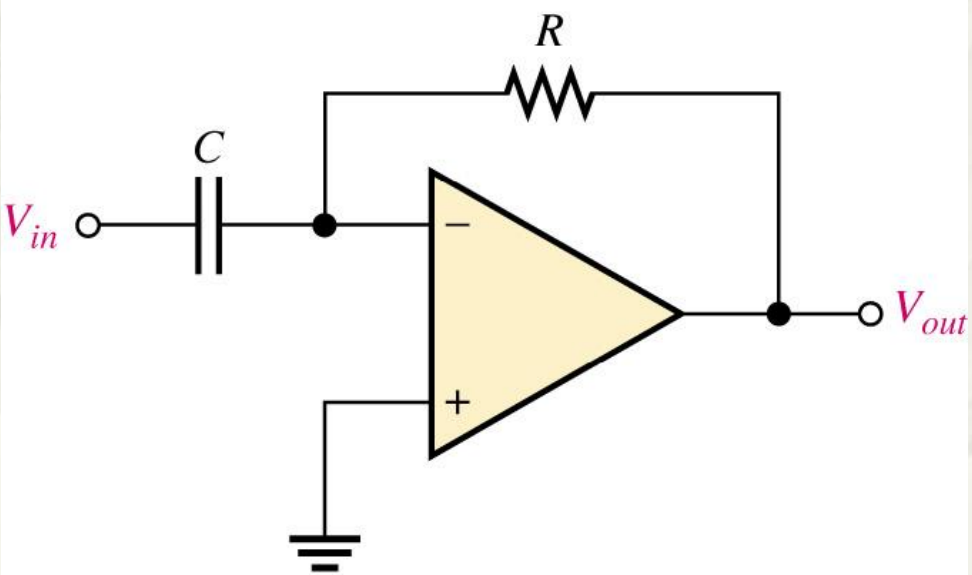
Solution



(b)

The Op-Amp Differentiator

- **Differentiator**
  - An differentiator is a circuit that produces an inverted output that approximates the rate of change of the input function.



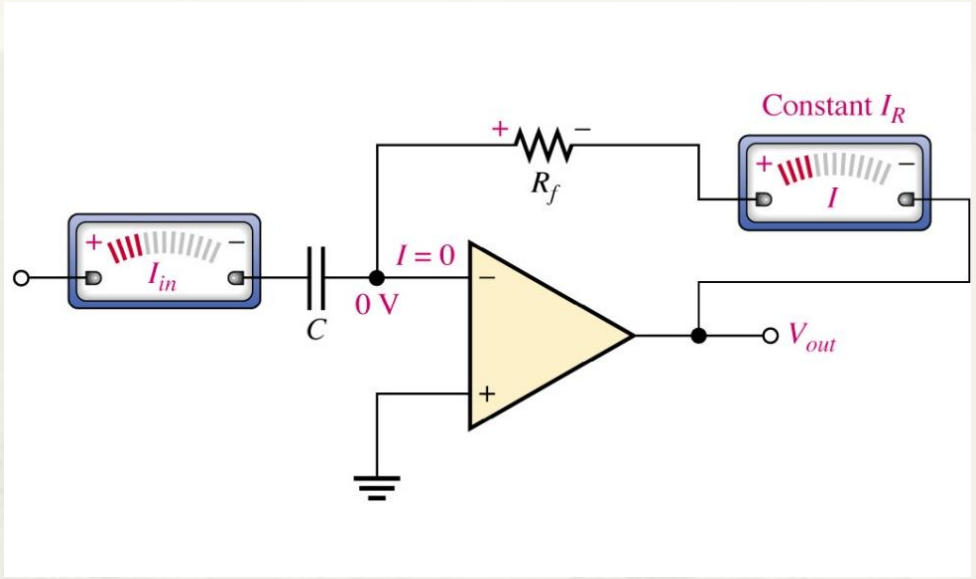
The Op-Amp Differentiator

- **Ideal differentiator**

$$V_C = \left(\frac{I_C}{C}\right)t \Rightarrow I_C = \left(\frac{V_C}{t}\right)C$$

$$V_{out} = -I_R R_f = -I_C R_f$$

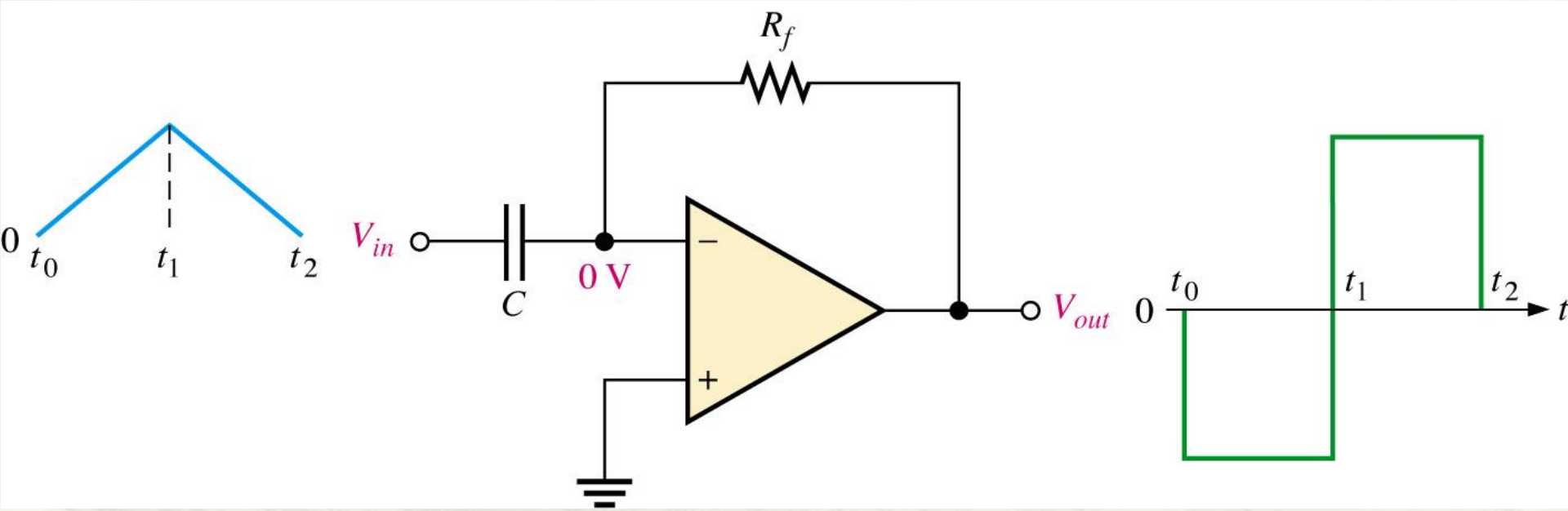
$$\Rightarrow \boxed{V_{out} = -\left(\frac{V_C}{t}\right)R_f C}$$



For instantaneous value:

$$\boxed{v_{out} = -R_f C \frac{dv_{in}}{dt}}$$

The Op-Amp Differentiator

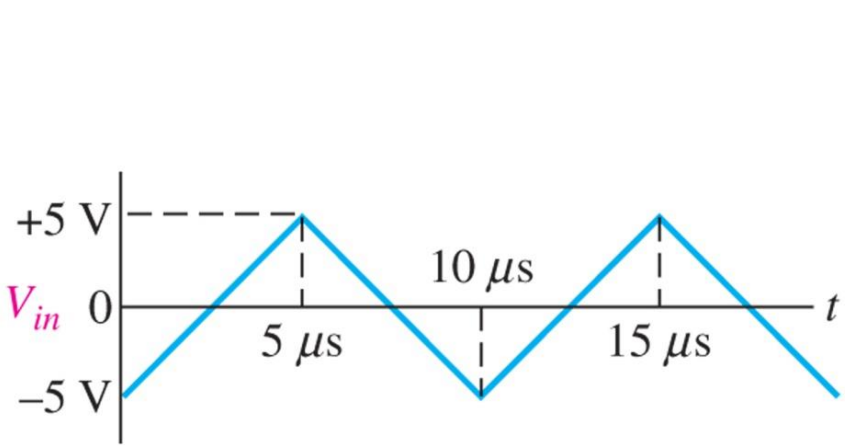


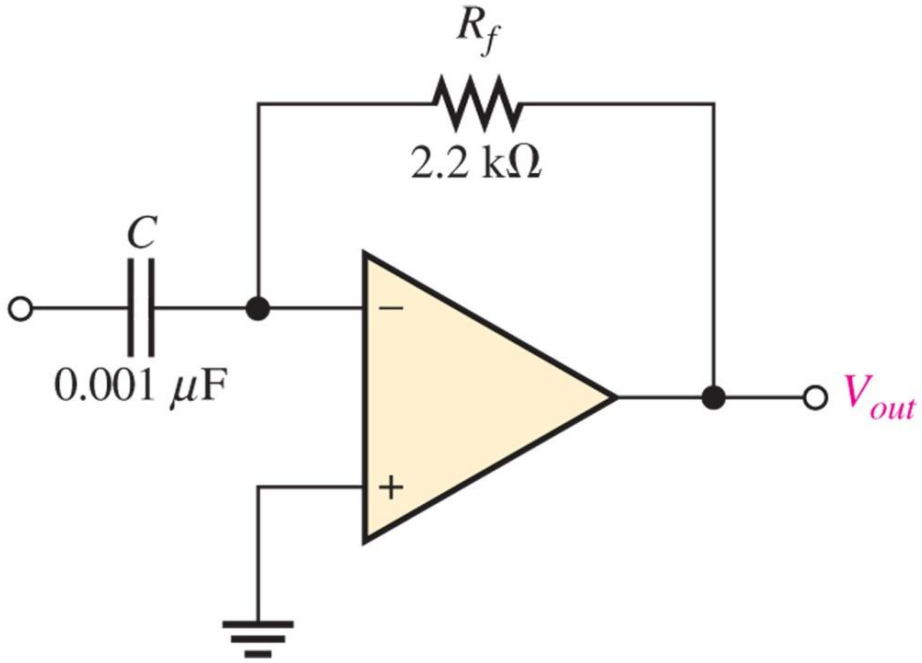
- **Rate of change**
  - The output voltage is proportional to the negative slope (rate of change) of the input.
  - The constant of proportionality is the time constant, $R_f C$

Example 7-7

Problem

Determine the output voltage of the ideal op-amp differentiator in Figure 7–26 for the triangular-wave input shown.





Example 7-7

Solution

Starting at $t = 0$, the input voltage is a positive-going ramp ranging from -5 V to $+5$ V (a $+10$ V change) in 5 μs. Then it changes to a negative-going ramp ranging from $+5$ V to -5 V (a -10 V change) in 5 μs.

Substituting into Equation 7–7, the output voltage for the positive-going ramp is

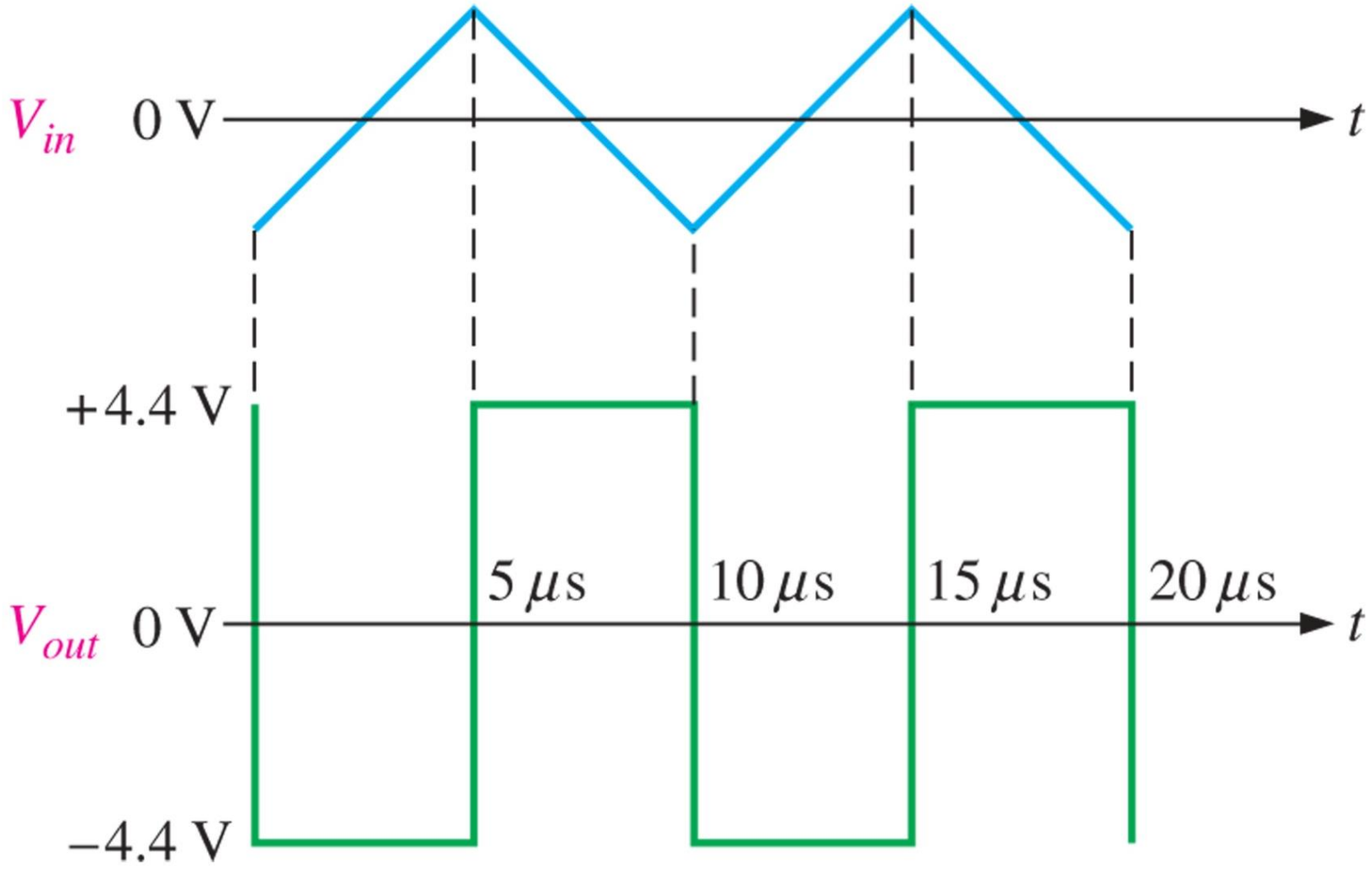
$$V_{out} = -\left(\frac{V_C}{t}\right)R_f C = -\left(\frac{10\text{ V}}{5\ \mu\text{s}}\right)(2.2\text{ k}\Omega)\,(0.001\ \mu\text{F}) = \mathbf{-4.4\text{ V}}$$

The output voltage for the negative-going ramp is calculated the same way.

$$V_{out} = -\left(\frac{V_C}{t}\right)R_f C = -\left(\frac{-10\text{ V}}{5\ \mu\text{s}}\right)(2.2\text{ k}\Omega)(0.001\ \mu\text{F}) = \mathbf{+4.4\text{ V}}$$

Example 7-7

Solution



7-4 Active Diode Circuits

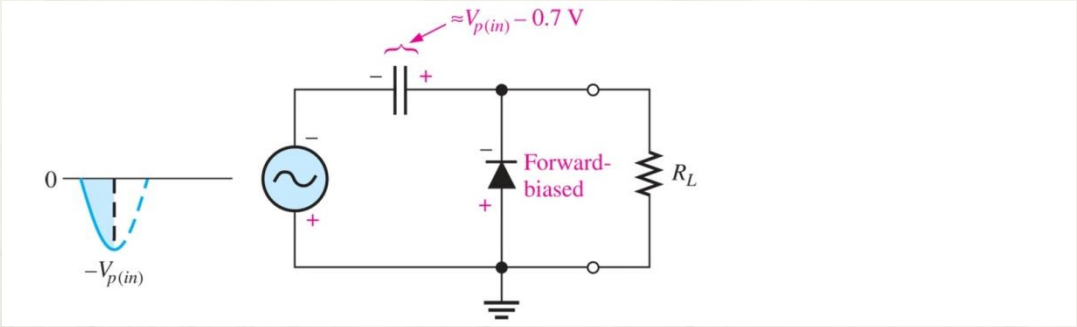
- Clamping Circuits
- Limiting Circuits
- Peak Detector

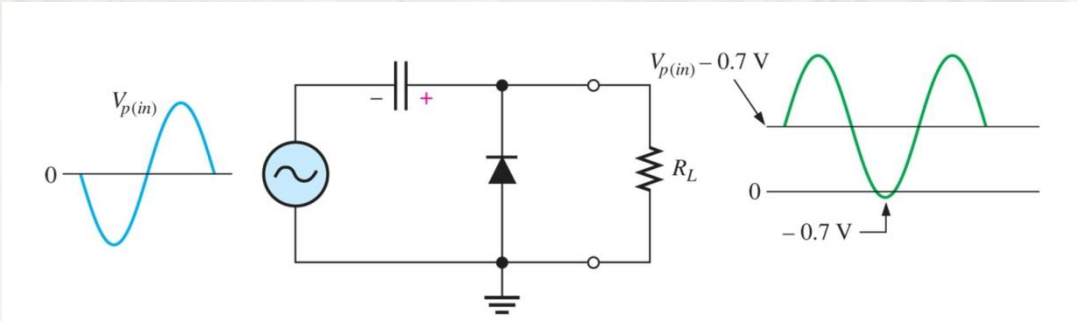
Clamping Circuits

- **Clamper**
  - A clamper is a circuit used to add a dc level to a signal voltage.
  - Clampers are often referred to as dc restorers because they are used to restore a dc level to a signal that has been processed through capacitively coupled amplifiers.

Clamping Circuits

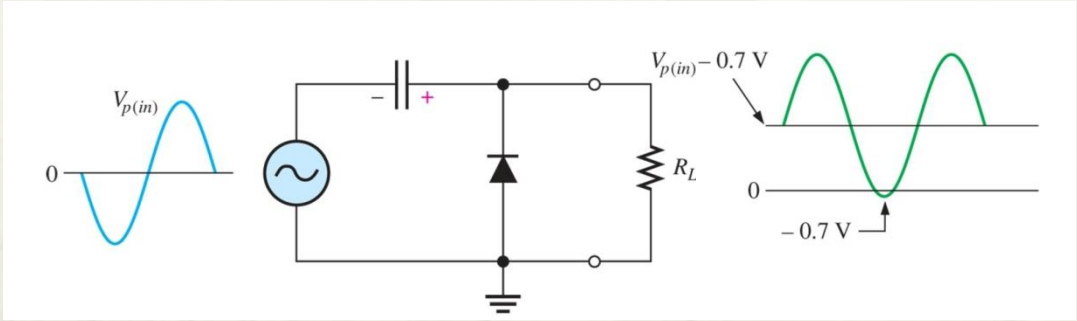
- **The basic Diode Clamper**
  - A simple passive diode clamping circuit
- **Positive clamping**
  - On the initial negative half-cycle
  - On the following cycles
  - The effection of *RC* time constant
  - The capacitor charge

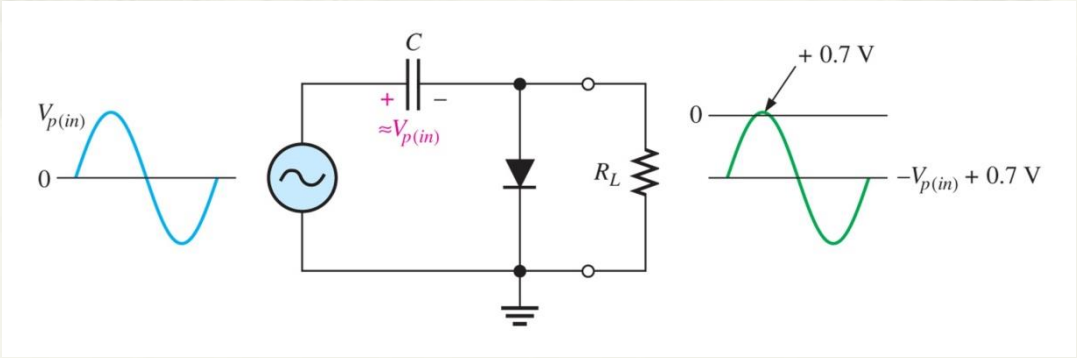




Clamping Circuits

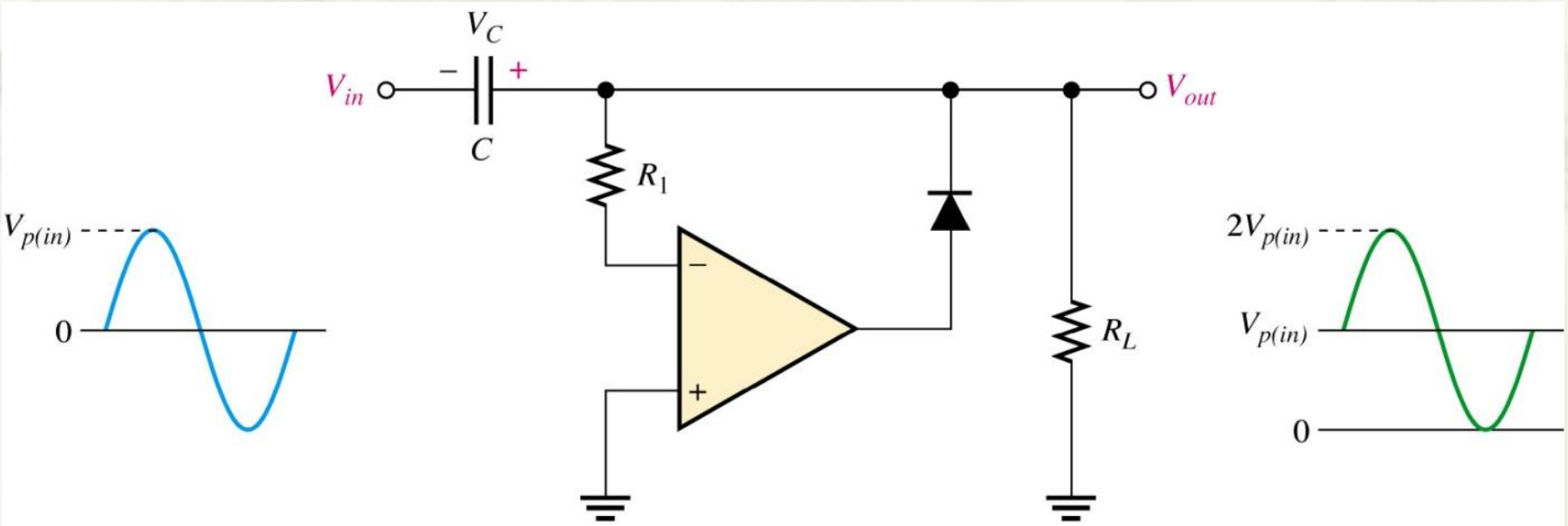
- **The basic Diode Clamper**
  - A simple passive diode clamping circuit
- **Positive clamping**
- **Negative clamping**





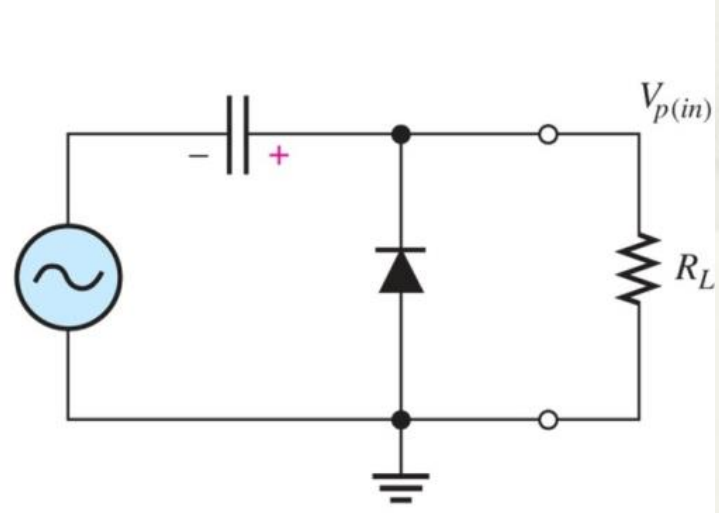
Clamping Circuits

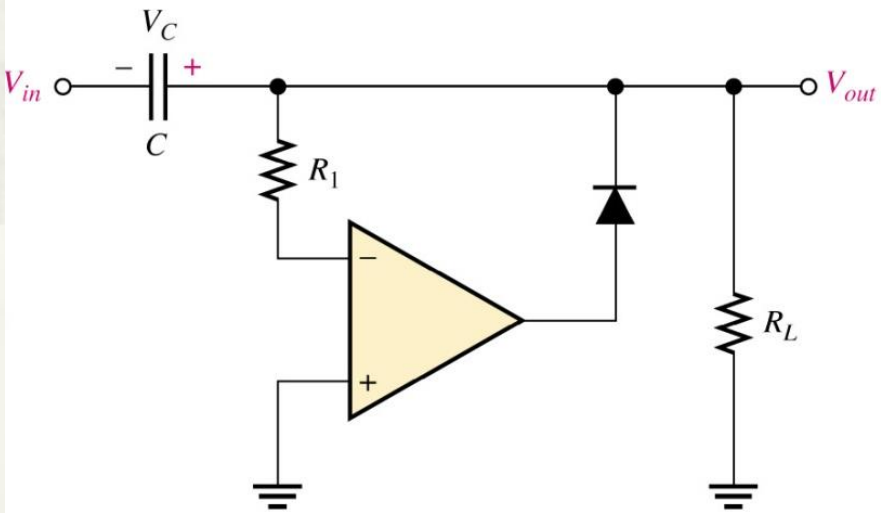
- **Active Clamping Circuit**
  - Active positive clamper
  - The operation
    - ✓ The first negative half-cycle
    - ✓ The following cycles



Clamping Circuits

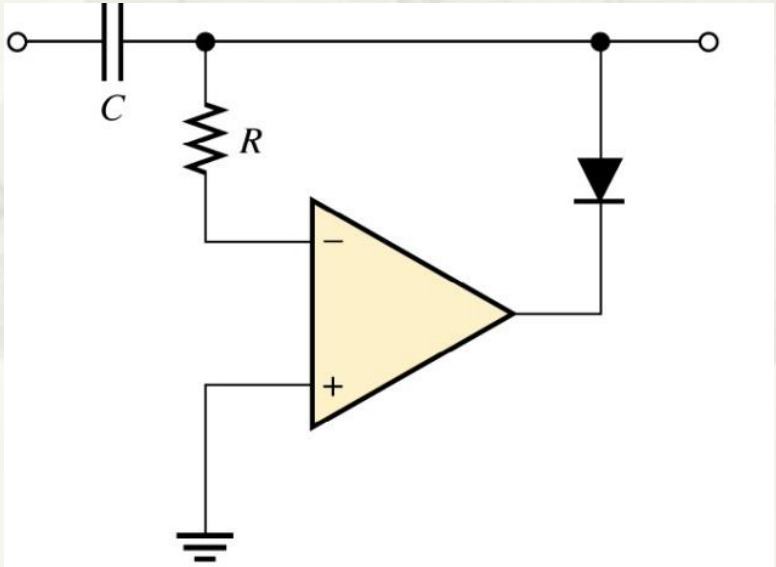
- **Active Clamping Circuit**
  - Advantages of active clamping circuit
    - ✓ The use of the op-amp eliminates the -0.7V peak found in the positive passive clamper output.
    - ✓ The use of the op-amp prevents loading the input source when the diode is forward-biased.



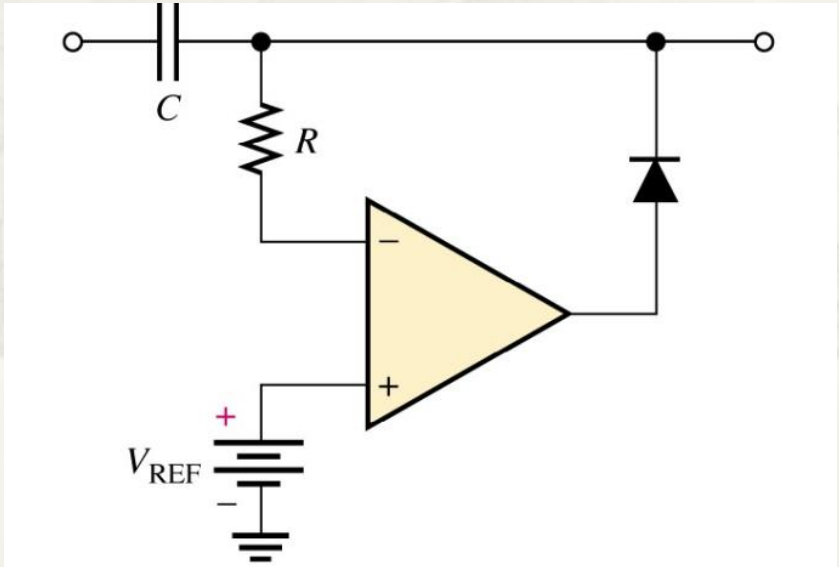


Clamping Circuits

- **Active Clamping Circuit**
  - Other active clamping circuits
    - ✓ Active negative clamper
    - ✓ Active positive clamper with nonzero reference



(a) Active negative clamper

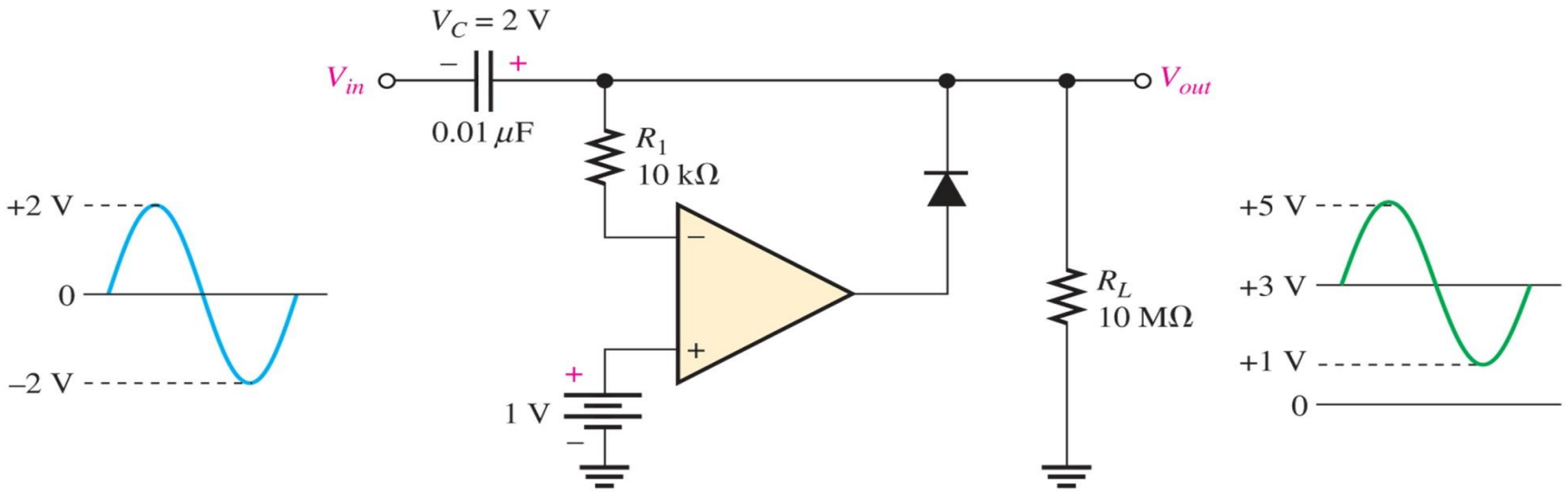


(b) Active positive clamper with nonzero reference

Example 7-8

Problem

Determine the output voltage for the clamping circuit in Figure 7–32 for the input voltage shown.

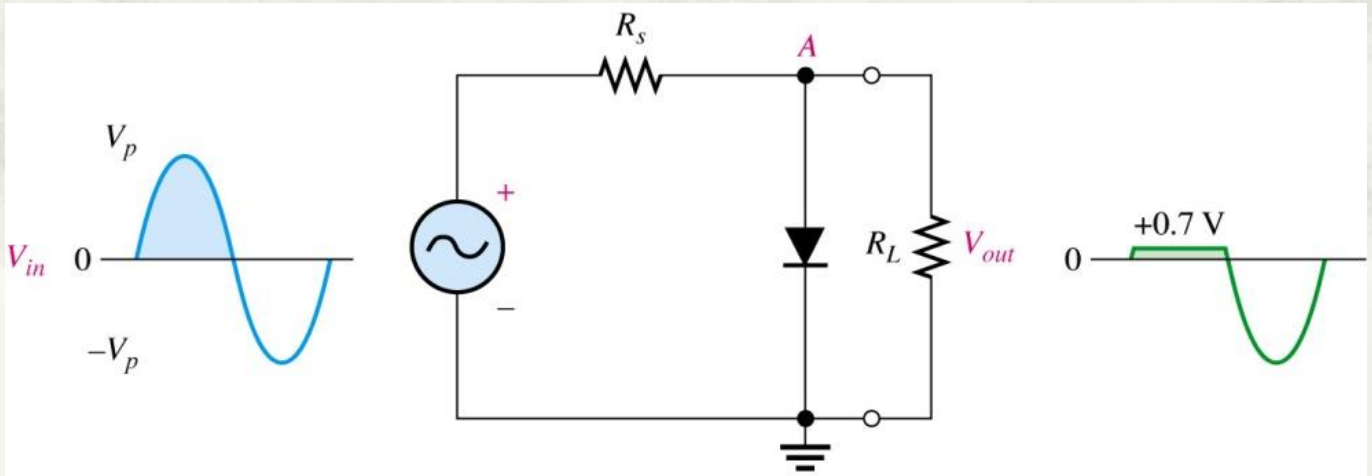


Solution

This is a positive clamping circuit and the reference voltage is +1 V, so the minimum peak value of the output voltage is also +1 V. The voltage is effectively shifted by 3 V, as indicated in the figure.

Limiting Circuits

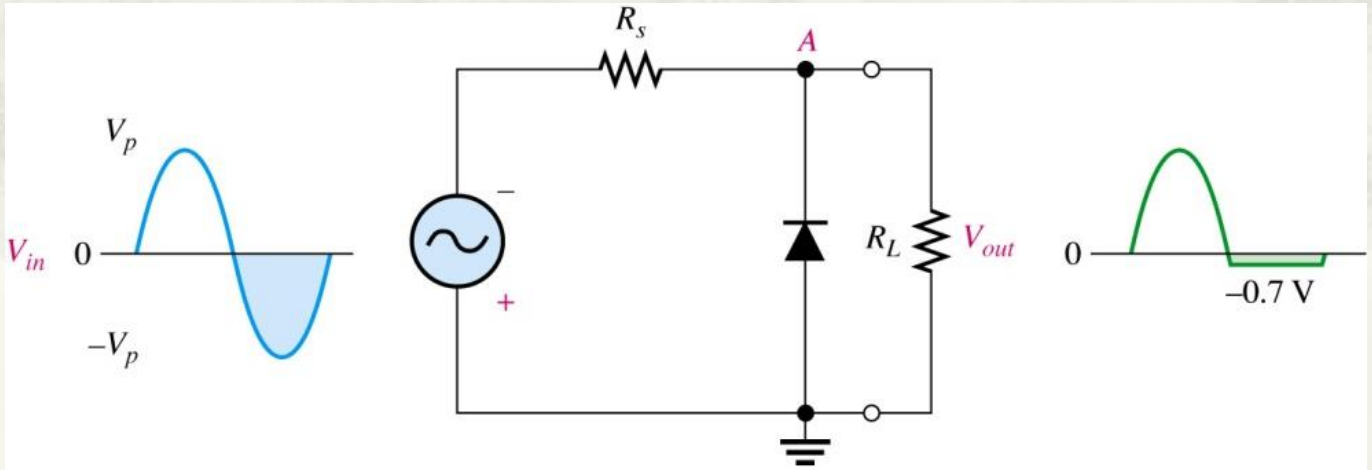
- **Limiter (clipper)**
  - Diode limiters cut off or limit voltage above or below specified voltage levels.
- **The Basic Diode Limiter**
  - Limiting of the positive alternation



(a) Limiting of the positive alternation

Limiting Circuits

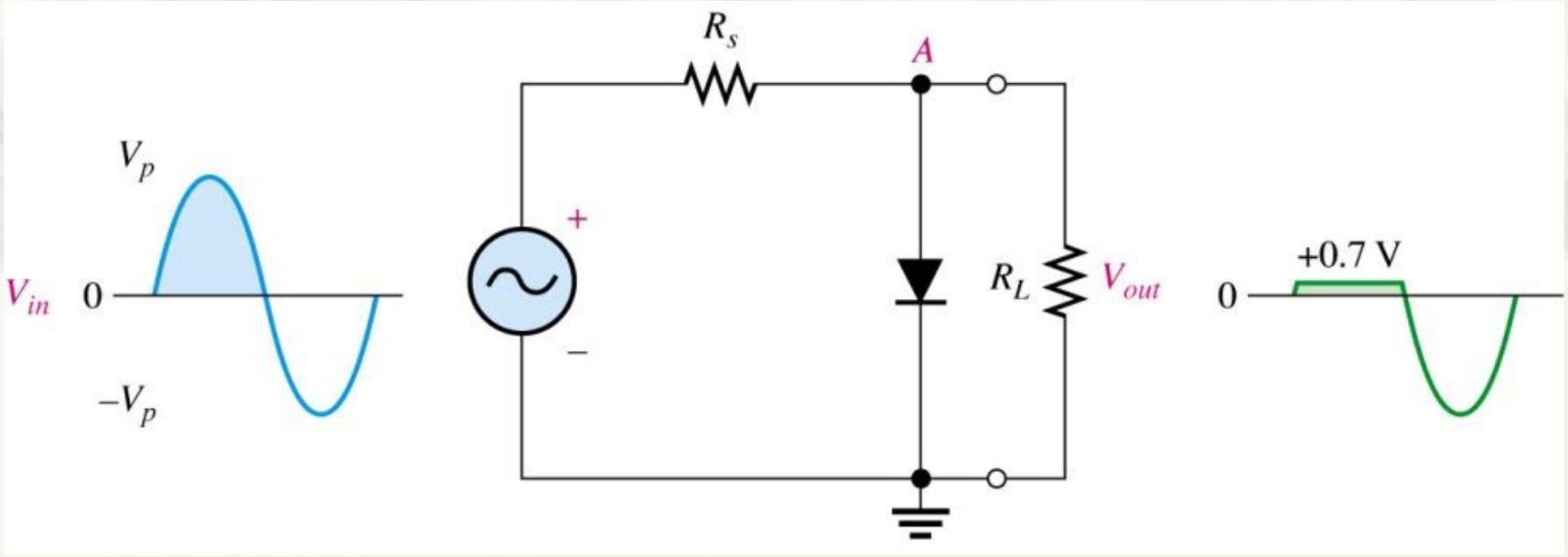
- **Limiter (clipper)**
  - Diode limiters cut off or limit voltage above or below specified voltage levels.
- **The Basic Diode Limiter**
  - Limiting of the positive alternation
  - Limiting of the negative alternation



(b) Limiting of the negative alternation

Limiting Circuits

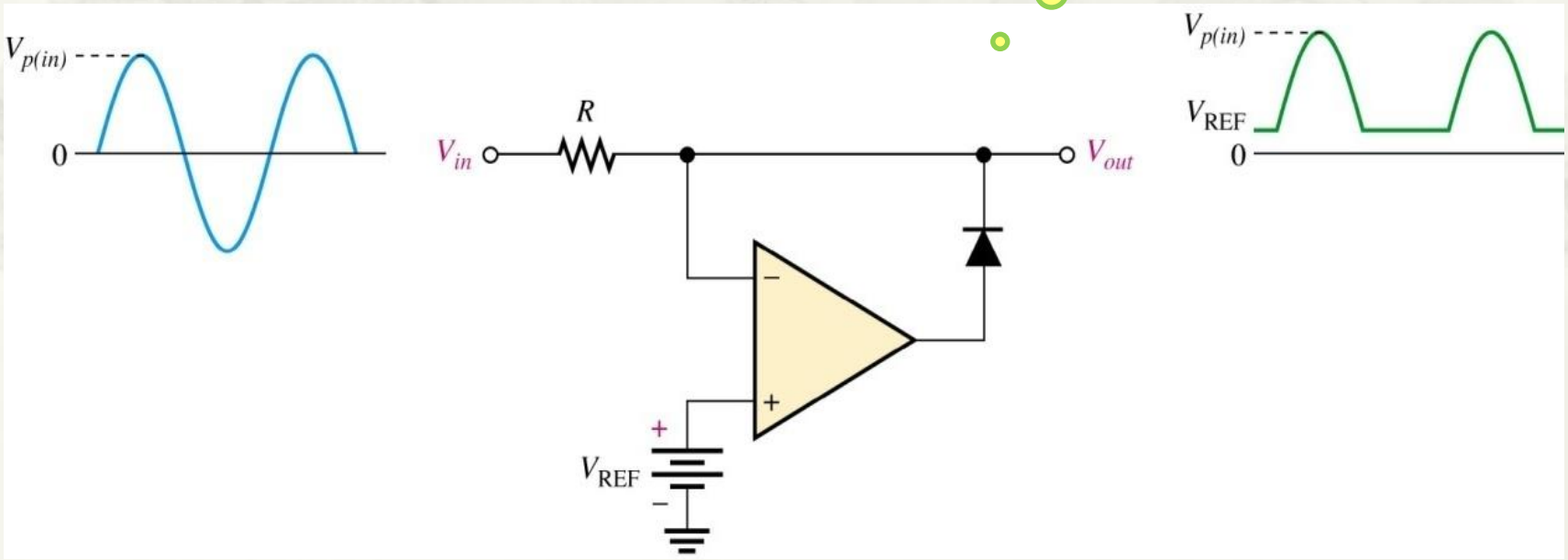
- **The Basic Diode Limiter**
  - To change the limiting level, a reference voltage source can be used in series with the diode or a zener diode can be used in place of the rectifier diode.
  - For good limiting action $R_L >> R_s$



Limiting Circuits

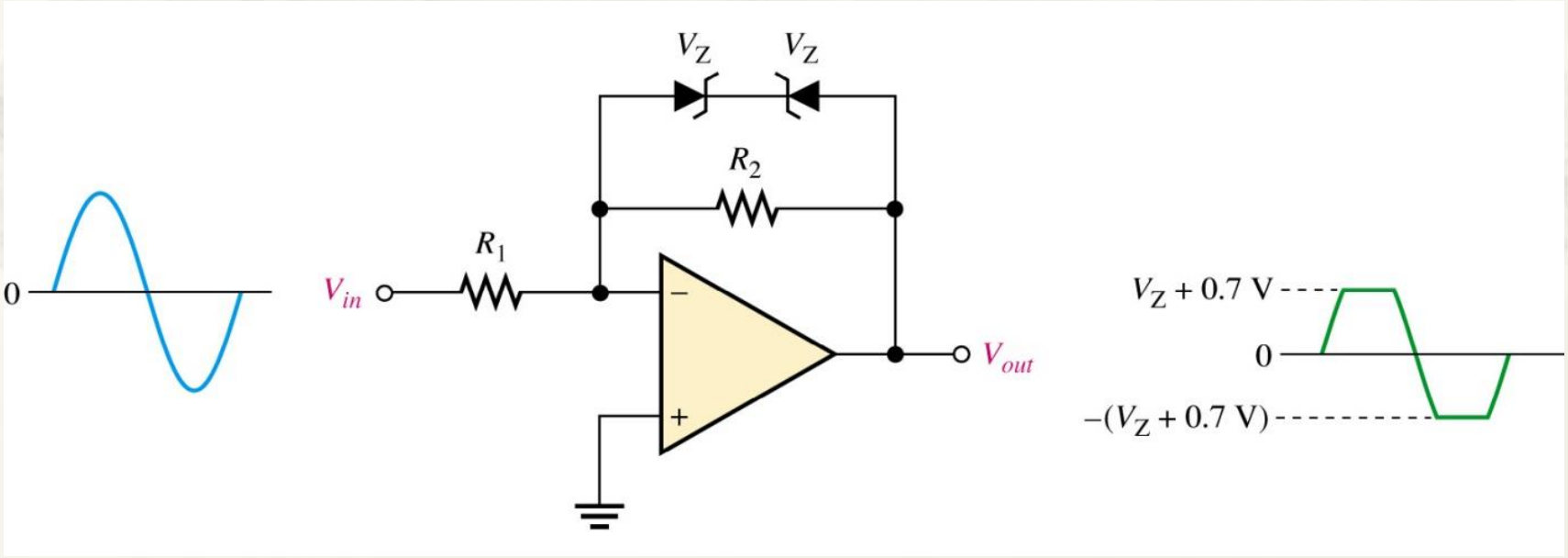
- **Active Limiting Circuits**
  - Op-amp limiting circuit that uses an op-amp and a diode
  - An active negative limiting circuit

What's the advantages compared to the basic diode limiters?



Limiting Circuits

- **Active Limiting Circuits**
  - Op-amp limiting circuit that uses an op-amp and a diode
  - An active negative limiting circuit
  - Limiting circuit using zener diodes in an inverting op-amp configuration

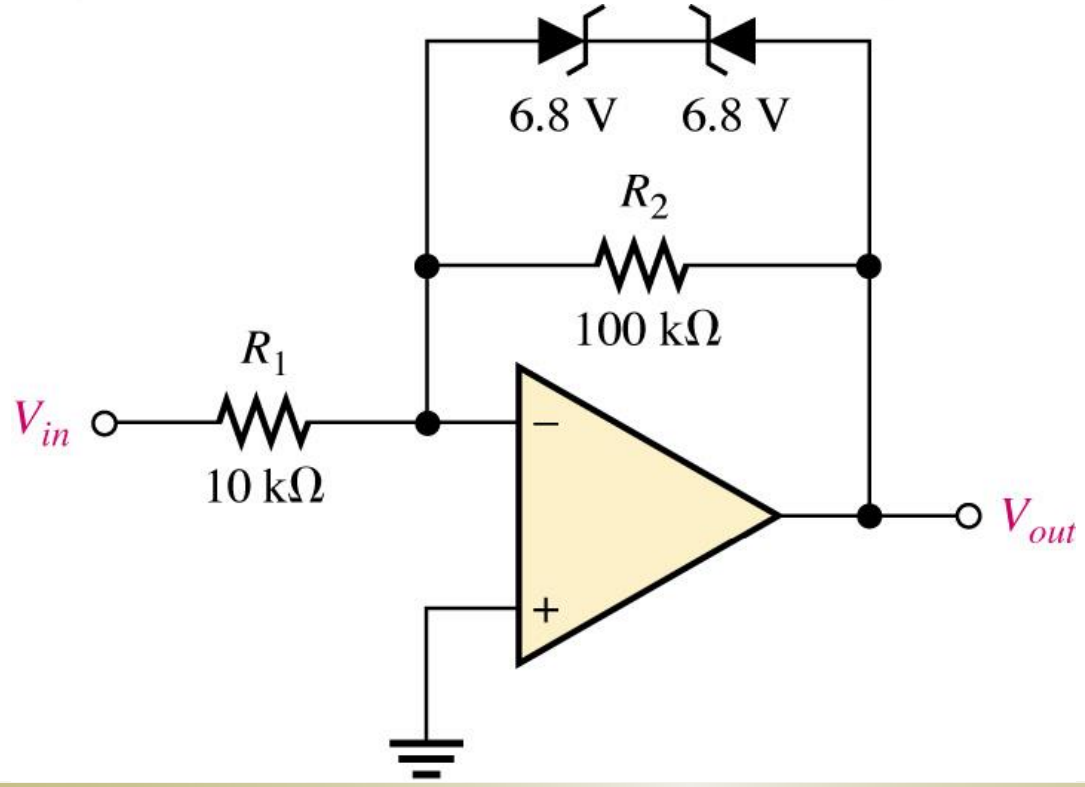


Example 7-9

Problem

(a) Describe the output waveform for the limiting circuit in Figure 7–37 for a 1 kHz sine wave with a peak value of 100 mV.

(b) Repeat part (a) for a 1 kHz sine wave with a peak value of 1 V.



Example 7-9

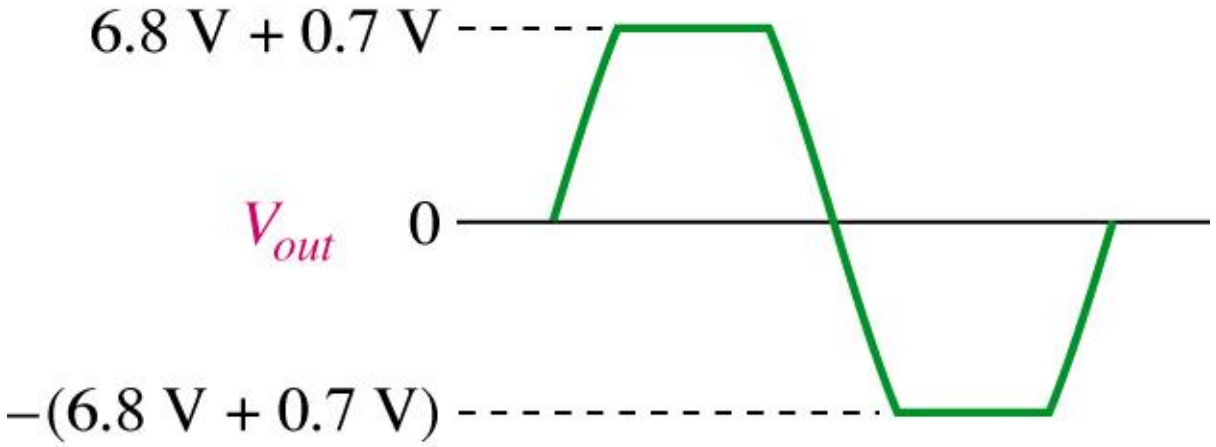
Solution

(a) The peak value of the output is

$$V_{p(out)} = (R_2/R_1)V_{p(in)} = (10)100\text{ mV} = 1\text{ V}$$

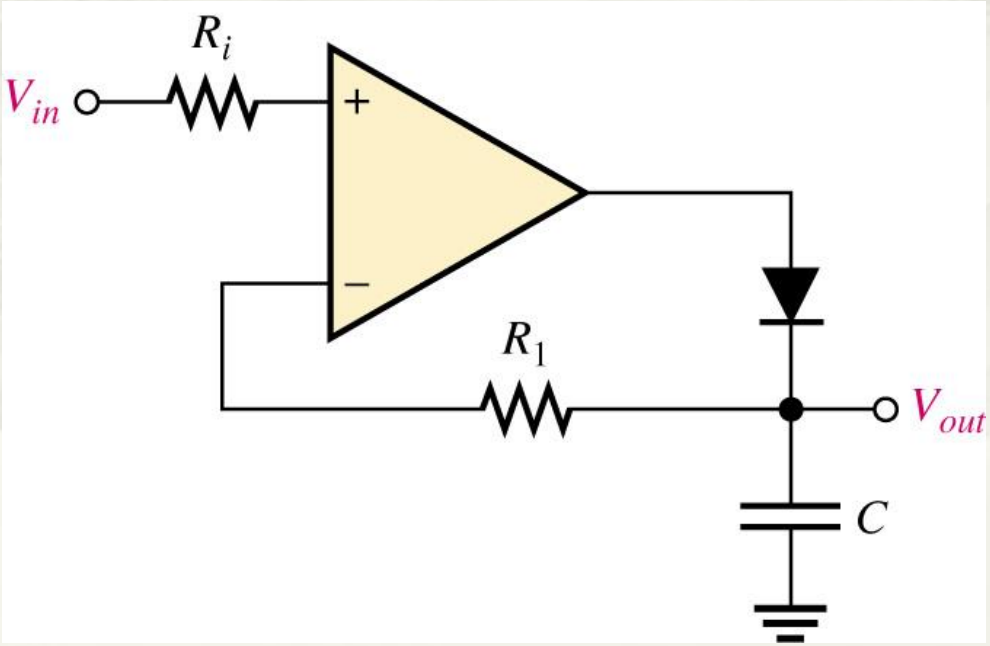
This value is less than the limiting level as set by the 6.8 V zener diodes, so the output is a sine wave that is not limited on its peaks.

(b) Since the peak output value exceeds the limiting level, the output is limited at ±(6.8 V + 0.7 V), as shown in Figure 7–38.



Peak Detector

- **Peak Detector**
  - A peak detector is a circuit used to detect the peak of the input voltage and store that peak voltage on a capacitor.

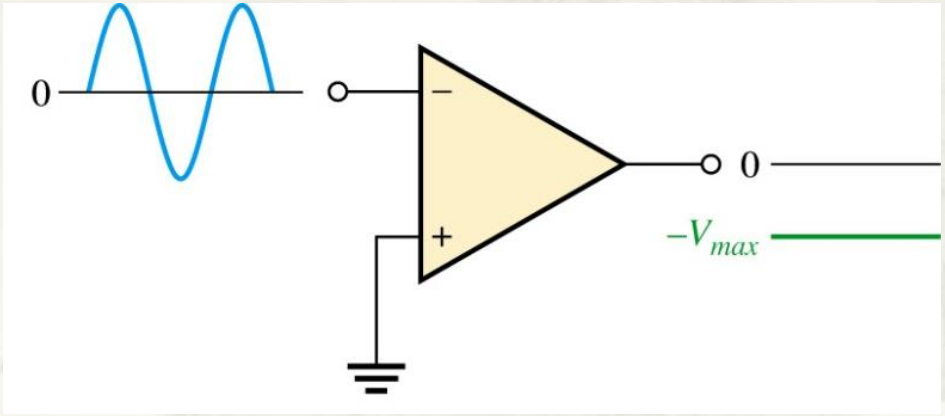


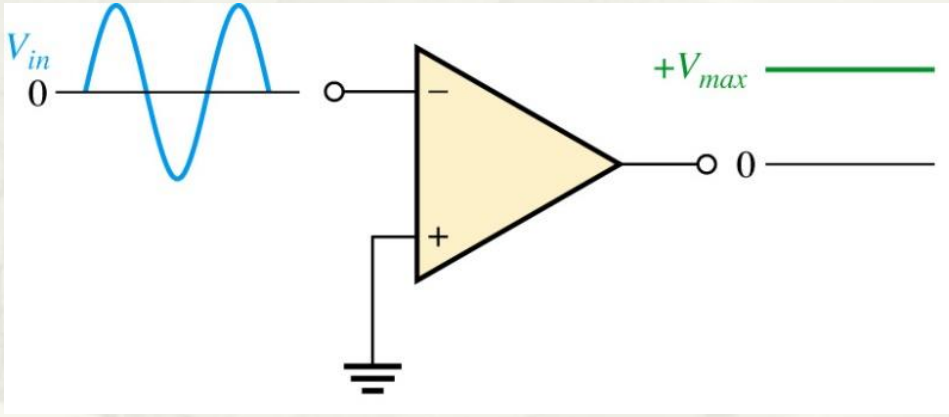
7-5 Troubleshooting

- Failure in Comparators
- Symptoms of Component Failure in Summing Amplifiers

Failure in Comparators

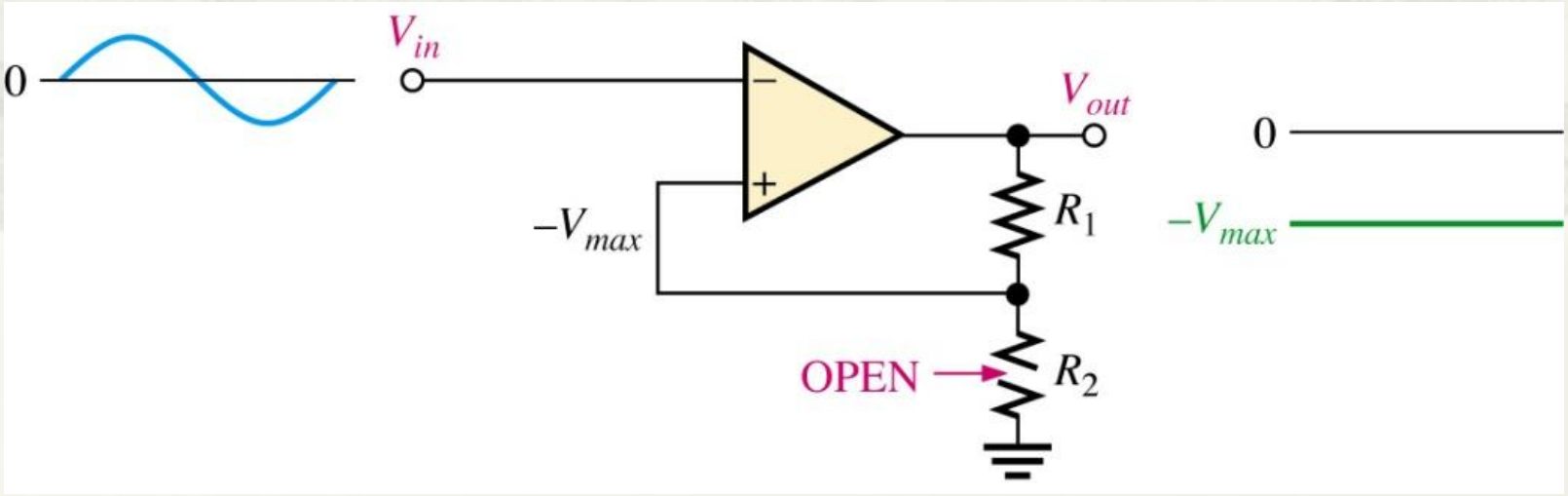
- **Internal comparator failures**
  - Typically result in the output being "stuck" in the HIGH or LOW state.





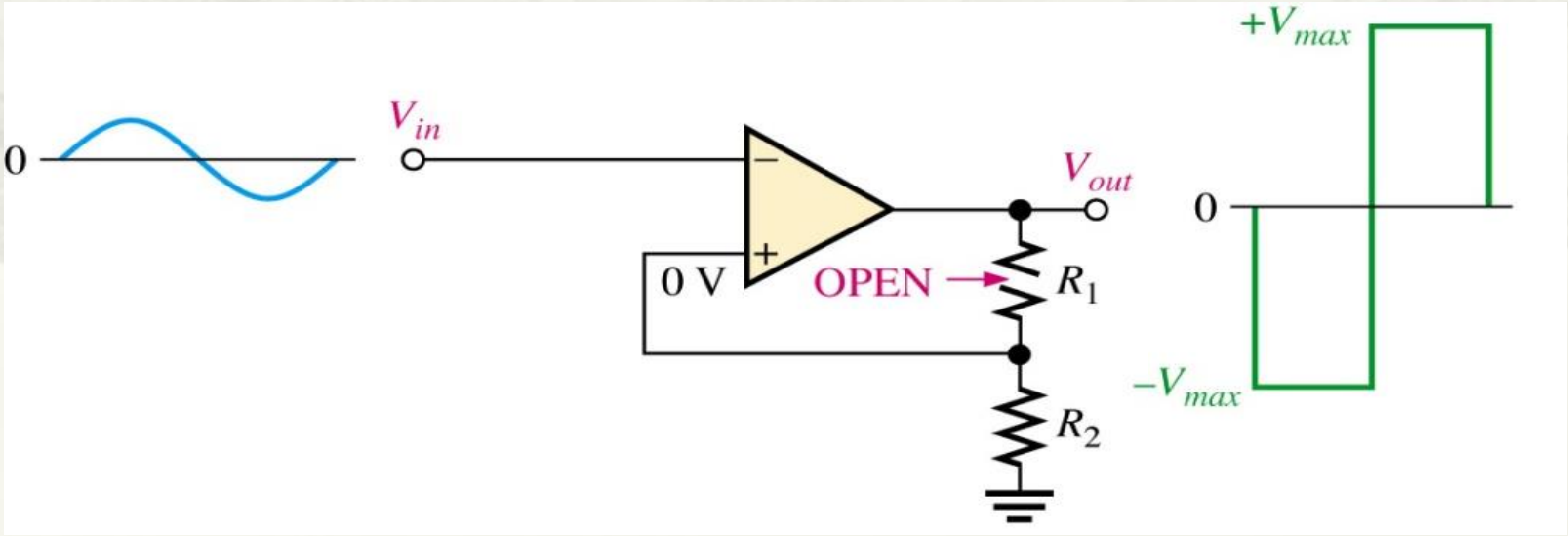
Failure in Comparators

- **Resistors open in comparator with hysteresis**
  - Open R_2 causes output to "stick" in one state (either positive or negative)



Failure in Comparators

- **Resistors open in comparator with hysteresis**
  - Open R_2 causes output to "stick" in one state (either positive or negative)
  - Open R_1 forces the circuit to operate as a zero-level detector



Symptoms of Component Failure in Summing Amplifiers

- **One of the input resistors open**
  - The output will be less than the normal value.
- **The feedback resistor opens**
  - The circuit will become a comparator.

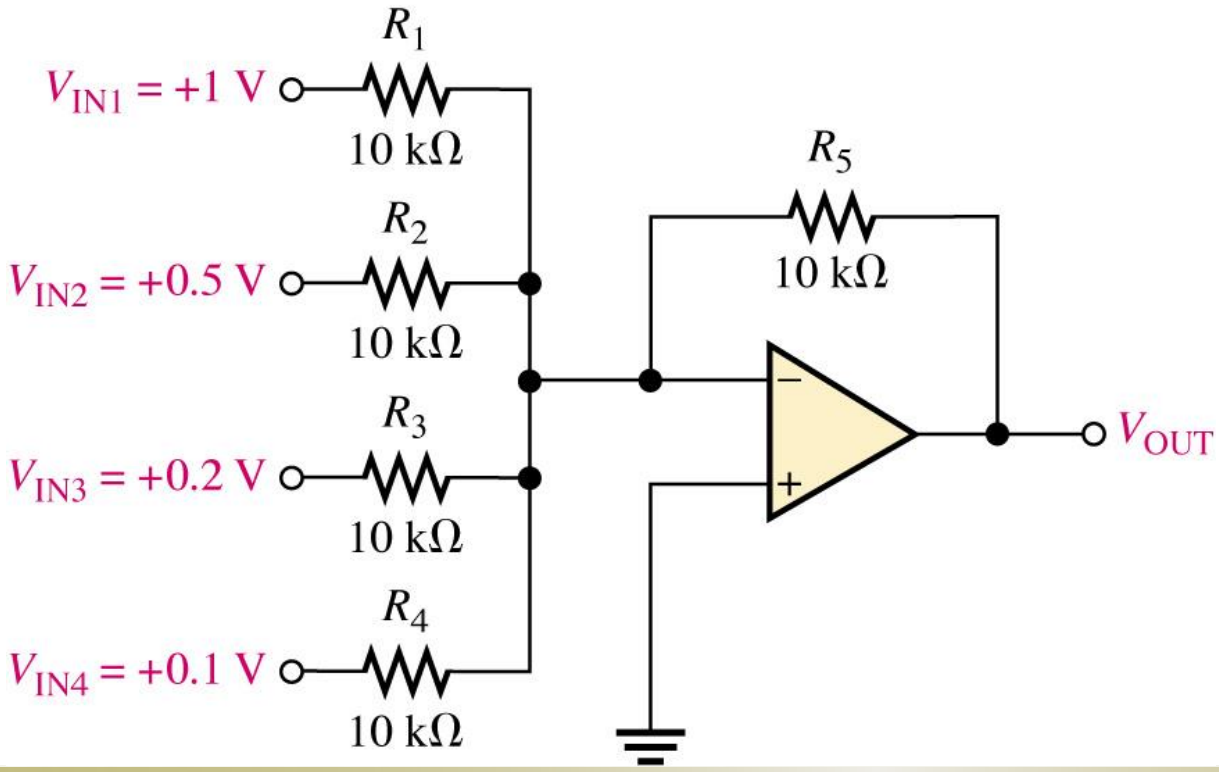
Example 7-10

Problem

(a) What is the normal output voltage in Figure 7–42?

(b) What is the output voltage if R_2 opens?

(c) What happens if R_5 opens?



Example 7-10

Solution

(a) $V_{OUT} = -(V_{IN1} + V_{IN2} + \cdots + V_{INn}) = -(1\,\text{V} + 0.5\,\text{V} + 0.2\,\text{V} + 0.1\,\text{V}) = \mathbf{-1.8\,V}$

(b) $V_{OUT} = -(1\,\text{V} + 0.2\,\text{V} + 0.1\,\text{V}) = \mathbf{-1.3\,V}$

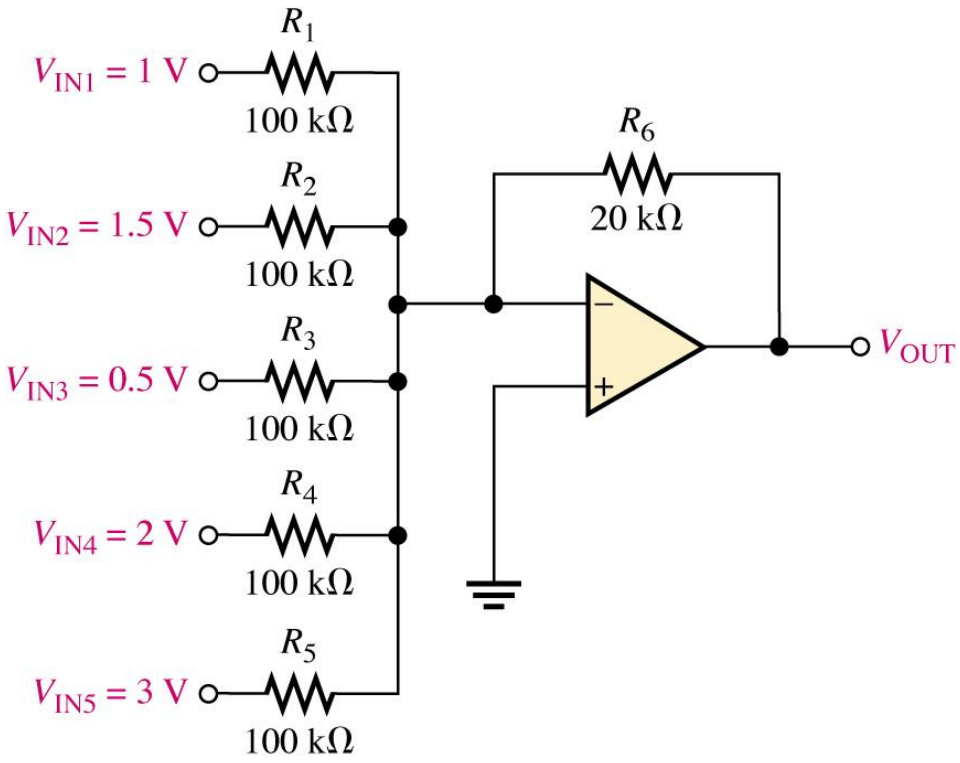
(c) If R_5 opens, the circuit becomes a comparator and the output goes to $-V_{max}$.

Example 7-11

Problem

(a) What is the normal output voltage for the averaging amplifier in Figure 7–43?

(b) If R_4 opens, what is the output voltage? What does the output voltage represent?



Example 7-11

Solution

Since the input resistors are equal, $R = 100\ \text{k}\Omega$. $R_f = R_6$.

$$\text{(a)}\quad V_{\text{OUT}} = -\frac{R_f}{R}(V_{\text{IN1}} + V_{\text{IN2}} + \cdots + V_{\text{IN}n})$$

$$= -\frac{20\ \text{k}\Omega}{100\ \text{k}\Omega}(1\,\text{V} + 1.5\,\text{V} + 0.5\,\text{V} + 2\,\text{V} + 3\,\text{V}) = -0.2(8\,\text{V}) = \mathbf{-1.6\ V}$$

$$\text{(b)}\quad V_{\text{OUT}} = -\frac{20\ \text{k}\Omega}{100\ \text{k}\Omega}(1\,\text{V} + 1.5\,\text{V} + 0.5\,\text{V} + 3\,\text{V}) = -0.2(6\,\text{V}) = \mathbf{-1.2\ V}$$

The 1.2 V result is the average of five voltages with the 2 V input replaced by 0 V. Notice that the output is not the average of the four remaining input voltages.

KEY OBJECTIVES

1. Describe the operation of several basic comparator circuits.
2. Describe the operation of several types of summing amplifiers.
3. Describe the operation of integrators and differentiators.
4. Describe the operation of clampers, limiters, and peak detectors.
5. Troubleshooting op-amp comparators and summing amplifiers.

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 1, 4, 7, 8, 9, 11, 12
- **Basic-plus Problems**
  - 18, 19, 20, 21, 22, 24, 25
- **Complementary Problems**
  - Analyze the relationship between the output voltage and the input voltages of a multi-input noninverting summer amplifier.